

Program overview

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Year 2023/2024
Organization Mechanical, Maritime and Materials Engineering
Education Master Offshore and Dredging Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master ODE 2023		Master Offshore & Dredging Engineering 2023					
MSc ODE Obligatory Courses (36 ECTS)							
CIEQ5002A	Structural Dynamics	4					
MT44021	Motions and Loading of Structures in Waves	5					
OE44030	Offshore Geotechnical Engineering	4					
OE44076	Offshore Renewable Energy	6					
OE44150	Marine Hydromechanics	5					
OE44155	Mechanical and Hydraulic Excavation	5					
OE44160	Limit States in ODE	2					
OE44165	Sediment Transport in Dredging Engineering	5					
MSc ODE Ethics and Engineering (choose 1 out of 2 (3 ECTS))							
WM0320TU	Ethics and Engineering	3					
WM0349WB	Philosophy of Engineering Science and Design	3					
MSC ODE Specializations							
MSc ODE Dredging Engineering (7 ECTS)							
OE44050	CFD for Dredging & Offshore Engineering	3					
OE44110	Subsea Engineering	4					
MSc ODE Structural Analysis & Design (15 ECTS)							
MT44010	Non-metallic materials in Marine Structures	5					
MT44085	Ultimate Strength of Marine Structures	5					
OE44085	Fatigue Strength of Marine Structures	5					
MSc ODE Offshore Renewable Energy (12 ECTS)							
OE44097	Offshore Support Structures	6					
OE44097 Toets 1	Exam Offshore Support Structures	2					
OE44097 Toets 2	Assignment Offshore Support Structures	4					
OE44100	Floating Structures & Offshore Moorings	6					
MSc ODE elective courses 1st year (courses from other specializations can be chosen as elective)							
AE4W31	Floating Offshore Wind Energy	3					
MT44001	Mechatronics in MT	5					
OE44090	Introduction to Computational Dynamics of Offshore Structures	4					
OE44105	Marine Pipelines	4					
MSc ODE 2nd year							
ODE Graduation Project (Obligatory 45 ECTS)							
OE54010-22	ODE Literature Research	10					
OE54035	ODE Thesis	35					
MSc ODE elective courses 2nd year (min. 15 ECTS)							
CIEM4302	Cold Regions Engineering	5					
MT44030	Torsion and Shear in Marine Structures	5					
MT44090	Fluid - Structure Interaction in Marine Structures	5					
OE44145	Research Exercise	6					
RO47001	Robot Dynamics & Control	5					
RO47002	Machine Learning for Robotics	5					
SC42001	Control System Design	5					
SC42015	Control Theory	6					
SC42056	Optimization for Systems and Control	3					
SET3055	Economics and Regulations of Sustainable Energy Systems	4					
TPM404B	Technology Entrepreneurship and Internationalization	5					
TUD4040	Joint Interdisciplinary Project	15					

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Master ODE 2023

Program Title	MSc Offshore & Dredging Engineering
Director of Education	Since 1926 Dredging Engineering and since 1975 Offshore Engineering courses are given at the Delft University of Technology. Offshore Engineering covers nearly everything what happens at the sea except for transport. In 2004 these two specialisations merged into the new MSc program Offshore & Dredging Engineering, a two-year (120 ECTS) curriculum leading to the MSc degree in Offshore & Dredging Engineering (ODE). Offshore & Dredging Engineering is a multidisciplinary cooperation between Civil Engineering, Mechanical Engineering and Maritime Technology. Students with those BSc degrees and BSc degrees in Aerospace Engineering and Applied Physics can directly enroll in this program. Students with a different background should first consult the director of education of Offshore & Dredging Engineering to explore the possibilities of enrollment. An appointment can be made by sending an email to the staffs secretary: A.C.M.deRuijt-Franke@tudelft.nl. This study guide is written as a guideline for students already enrolled in the MSc program Offshore & Dredging Engineering and for students who consider choosing Offshore & Dredging Engineering to obtain their masters degree. It concentrates on the academic content of the curriculum and it presents practical information regarding the organization of the curriculum. Since information, in particular the class schedules, may change, the latest information can be found on our website: https://www.tudelft.nl/onderwijs/opleidingen/masters/ode/msc-offshore-dredging-engineering/ We wish you a very successful stay in our Offshore group and with your study. Your success will depend to a large extent on how much energy you are willing to put in being an active participant in Offshore & Dredging Engineering. Dr.ir. S.A. Miedema, Director of Education
Introduction 1	Students with TUD Bachelor degree Wb, MT, LR, CT and TN can be admitted to the masterprogramme without following a bridging programme.
Program Goals	The Delft Offshore Curriculum Objectives All TU Delft MSc curricula last nominally two academic years. Within these two years - by building upon a participants BSc background the overall Offshore & Dredging Engineering curriculum objective is to prepare successful participants for an active role in society that is related in some way to the offshore or dredging equipment industry - especially when this latter equipment is to be used in exposed or deep water locations. After completion of the curriculum one can now work with new and more complex problems (probably only within ones specialty area) and develop, evaluate and document solutions for them. This defines the objective for the Offshore & Dredging Engineering MSc Degree curriculum.
Exit Qualifications	Designing ones Individual Program Many beginning MSc participants are concerned and insecure about what seems like a myriad of decisions that they feel they have to make about their own specific program choices (which courses to include) without sufficient information. Indeed, nearly everyone arrives at the doors of the Offshore & Dredging Engineering curriculum without bringing any specific preparation from their BSc program experience. It will be refreshing for these new arrivals to know that most of their program decisions can be delayed at least until they (individually) have had a few weeks to become orientated within Offshore & Dredging Engineering; this reduces the pressure and gives time to develop a better motivation for making choices for the individual study program. One will find, for example, that only one ODE course taught in the first quarter of the first MSc year is associated with a specific specialisation. Feel free as well to consult with the ODE curriculum leader about professional objectives and program choices. He can be reached with a question or for an appointment via an e-mail to A.C.M.deRuijt-Franke@tudelft.nl.
Program Structure 1	Curriculum Structure Now that the curriculum has been described in very general terms and the requirements for admission have been defined as well, attention in this chapter can switch to the more detailed description of the courses which make up the Offshore & Dredging Engineering MSc curriculum. One is reminded that the entire MSc curriculum lasts two academic years, which amounts to a total of 120 ECTS (European Credit Transfer System) of work. Generally speaking, each participant is expected to include the following in their personal study program: - The core curriculum of courses of Offshore & Dredging Engineering - One specialisation - Complemented with electives to at least 120 ECTS credits. As mentioned one of the five specialisations has to be chosen: - Bottom Founded Structures - Dredging Engineering - Floating Offshore Structures - Ship & Offshore Structures - Offshore Renewable Energy

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MSc ODE Obligatory Courses (36 ECTS)

Introduction 1	This section outlines course requirements that are more or less universal - independent of ones further choices. The knowledge and skill conveyed by these activities forms the heart of the offshore engineering curriculum; nearly all of the subsequent offshore engineering courses build upon these. These courses are coloured green on the Offshore Engineering MSc Curriculum Diagram which may be downloaded from the website www.offshore.tudelft.nl. Detailed information on the core and specialisation courses is provided in the overview of courses. Also contact details of the teachers are given there.
Introduction 2	In below mentioned table all courses are mandatory with the following exception: * AES1730 is mandatory for students with a BSc Mechanical Engineering background * SC4026 or WB2105MT is mandatory for students with a BSc Civil Engineering background

CIEQ5002A	Structural Dynamics	4
Responsible Instructor	Prof.dr. A. Metrikine	
Responsible Instructor	Dr.ir. H.R. Schipper	
Responsible Instructor	A. Tsouvalas	
Responsible Instructor	Dr. M. Pavlovic	
Contact Hours / Week x/x/x/x	This equivalency course replaces the following course from the old master programme: CIE4140. The timetable is identical to that of the (unit 2 from) module CIEM5000 in the new programme. Only lectures related to "Structural Dynamics" in unit 2 need to be attended. Please see the study guide description of CIEM5000 and the information below. Please contact the module manager at the start of the module for registration and additional information.	
Education Period	2 3	
Start Education	2 3	
Exam Period	2 3 4	
Course Language	English	
Course Contents	CIEQ5002A replaces CIE4140 Structural Dynamics, the latter course being discontinued with the introduction of the new MSc Curriculum Civil Engineering in September 2022. CIEQ5002A contains a subset of the learning objectives of the larger module CIEM5000 Structural Engineering Track Base (15 ECTS), all taken from Learning Unit 2 "Mechanics of Slender Structures" from both the 2nd and 3rd educational period (Q2 and Q3). Please notify the lecturer Dr. Tsouvalas in advance in case you take this course as student from the old CE-Programme, in order to allow for correct administration of your grade.	
Study Goals	Unit 2 of CIEM5000 treats the mechanics and dynamics of slender structures. In Q2, simple cases are treated in which analytical solutions of single elements are possible. The goal is to explain physics and the basics of the methods. In Q3, the focus is shifted towards more design-type assignments in which the students apply the solution methods in more complex systems consisting of combined one-dimensional structural elements and plates together with complex boundary/interface conditions. The unit covers: - Formulation of governing equations of one-dimensional and two-dimensional slender structures. - Statics and dynamics of cables, rods, shear beams and beams in bending. - Statics and dynamics of plates in bending. - Statics and dynamics of combinations of one-dimensional slender structures. - Statics and dynamics of plates undergoing in-plane deformations. - Statics and dynamics of combinations of one-dimensional and two-dimensional slender structures. After completion of this unit, students will be able to (learning objectives from CIEM5000): 6. Formulate the governing equations of motion of a class of one-dimensional slender structures (beams, cables, rods) and plates subjected to static and dynamic loads in- and out- of plane. 7. Analyse 1D slender structures subjected to static loads and dynamic loads and evaluate the structural response. 8. Analyse plates subjected to static and dynamic loads in- and out- of plane and evaluate the structural response. 9. Create mathematical models of complex structures consisting of one-dimensional elements and plates subjected to static and dynamic loads and evaluate their structural response. To fulfil the equivalent course CIEQ5002A, only a subset of the above topics and learning objectives focussing on dynamics will have to be studied.	
Education Method	Lectures. Practical sessions. For more information please check the Brightspace page of the module CIEM5000.	
Assessment	Please notify the lecturer Dr. Tsouvalas in advance in case you take this course as student from the old CE-Programme, in order to allow for correct administration of your grade. Material delivered in Q2 will be assessed on the basis of a written exam. In Q3, the assessment will be on the basis of a final assignment, focusing on the dynamics aspects. The weight for the assessments above are as follows: - Mechanics of Slender Structures - Written Exam (Q2) 60% - Mechanics of Slender Structures - Report (Q3) 40% For the equivalent course CIEQ5002A, the focus will be on dynamics.	
Expected prior Knowledge	see CIEM5000	
Academic Skills	see CIEM5000	
Literature & Study Materials	see CIEM5000	
Judgement	see assessment	
Permitted Materials during Exam	see CIEM5000	
Collegerama	Yes	

MT44021	Motions and Loading of Structures in Waves	5
Responsible Instructor	Dr.ir. P. Naaijen	
Instructor	Dr.ir. P.R. Wellens	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	Motions and loading of structures is about the following topics. Part 1: Part 1 covers theory of linear excitation and linear motion response of floating structures in waves. Students learn how the linear motion transfer function (RAO) relating ship motion response to incoming wave is computed, including the numerical approach involved and how workability in a given wave spectrum can be assessed including extreme value statistics. In Part 2: Higher order potential theory and inclusion of non-linear effects in ship motions. Applications to motion of moored ships and to the determination of workability. The use of Computational Fluid Dynamics (CFD) for the determination of extreme loads on moored FPSO in extreme seas. One exercise accompanies this part.	
Course Contents	Floating structures: wave forces & motions, nonlinear problems, applications Participants completing this segment successfully will have a superior knowledge of and be able to predict the motion of floating bodies in the sea. They will be familiar with first order ship motions in irregular waves as well as drift forces, resulting from nonlinear phenomena. They can also apply this to applications such as station keeping and the determination of offshore workability. Part 1, instructed by Peter Naaijen, is about linear forces, motions and about potential coefficients. Part 2, instructed by Peter Wellens, is about nonlinear forces and the resulting motions.	
Study Goals	Participants who have successfully completed the course will be able to carry out computations at a superior knowledge level involving: - linear forces on and resulting motions of ships - nonlinear forces on and resulting ships motions - workability prediction.	
Education Method	Lectures, exercises and assistance sessions for exercises	
Literature and Study Materials	For Part 1, separate lecture notes by Peter Naaijen are available. As an additional reference the Lecture notes "Offshore Hydromechanics" by Journée, Massie and Huijsmans can be used. Exercises "Offshore Hydromechanics" by Journée and Massie.	
Prerequisites	All participants are required to have successfully completed a basic university level courses in dynamics (mass-spring-damper systems etc.), fluid mechanics and free surface waves before starting this course.	
Assessment	Exams: Both part 1 and part 2 provide a summative assessment (exercise) which is not graded. One graded written exam (summative) for Part 1 and Part 2 together at the end of the course. (In case on site examination will not be possible due to COVID19 related circumstances, online examination will be arranged.)	
Remarks	Aggregation of old courses MTM1410 and OE4630 part 2 and 3	
Department	3mE Department Maritime & Transport Technology	

OE44030	Offshore Geotechnical Engineering	4
Responsible Instructor	Dr. E. Kementzetzidis	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course addresses relevant topics in Soil Mechanics and Geotechnical Engineering related to offshore energy production (for both oil&gas developments and renewable sources). The core subject is the analysis and design of the most common foundation systems for offshore structures. After some preliminary recaps on soil behaviour, the response of soils to environmental cyclic loading is illustrated and discussed. Then, essential concepts about marine site investigation are discussed. The course is brought to conclusion by widely address its core topics, concerning the analysis/design of both shallow and deep offshore foundations.	
Study Goals	At the end of this course you will be able to: 1. recognise and describe the main features of offshore soil behaviour under environmental loading; 2. describe the principles for planning offshore site surveys for soil characterisation purposes; 3. analyse/design the main shallow foundation systems according to standard analytical approaches 4. analyse/design offshore piles according to standard analytical approaches	
Education Method	A combination of pre-recorded (on-line) and live (in-class) sessions. Both fundamentals and practical applications will be covered in both teaching environments.	
Assessment	Assessment is fully through a written, open-book exam at the end of the education period. A similar resit will take place at the end of the following education period. In case of no access to the faculty allowed, the written exam will be organised remotely - with specific checks on students' conduct and plagiarism.	
Remarks		
Department	3mE Department Maritime & Transport Technology	
Contact	For more information please contact the course coordinator by email (f.pisano@tudelft.nl)	

OE44076	Offshore Renewable Energy	6
Responsible Instructor	Dr.ir. A. Jarquin Laguna	
Instructor	Ir. A.C.M. van der Stap	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This course will introduce the students to the technical aspects and detailed understanding of ocean energy technologies (including floating wind, wave, and tidal current). The course combines lectures on offshore renewable energy technologies, CO2 and energy storage. The students will participate in a realistic project teamwork with young industry coaches and working on actual cases with stakeholder and risk management assessment, modelling and energy extraction analysis from the offshore environment, including an overview of the project development of safe and reliable offshore energy technologies.	
Study Goals	by successfully completing the course you should be able to: Explain and describe the physical principles of energy conversion for wave energy and tidal stream . Compare different marine renewable energy sources based on calculations of their energy content. A few examples of such sources include offshore wind, ocean waves, tidal currents Identify and evaluate the design considerations and technical challenges of offshore wind, wave and tidal energy technologies. Generate future-proof solutions for offshore energy projects with a focus on minimization of the CO2 footprint. Define technical, political, contractual, economic and operational constraints imposed on industrial projects throughout its entire life cycle. Explain the crucial importance of uncertainties and economics in the decision process in the course of developing a project. Apply basic project analysis and management techniques. Accommodate conflicting requirements in an offshore design environment.	
Education Method	Lectures and project teamwork with industry coaches.	
Assessment	Grading based on a group assignment.	
Department	3mE Department Maritime & Transport Technology	

OE44150	Marine Hydromechanics	5
Responsible Instructor	Dr. H. Seyffert	
Contact Hours / Week x/x/x/x	HC 4.0.0.0 W 2.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The main topics of this course are: - hydrostatics / floating stability - constant potential flow - constant real (viscous) flow - free surface (regular & irregular) waves - wave spectra - wave forces on slender structures	
Study Goals	After this course, students should be able to: - Identify the relevant hydromechanical principles of the ocean environment for the analysis of offshore systems - Formulate the hydromechanical principles of the ocean environment for the analysis of offshore systems - Determine the hydromechanical loading on fixed rigid offshore systems - Assess the implications of simplifications of hydromechanical principles and corresponding assumptions on the analysis of fixed rigid offshore systems	
Education Method	classes, practice workshops	
Literature and Study Materials	Offshore hydromechanics, third edition, by Massie, Journée and Huijsmans Handout for wave forces on slender structures	
Assessment	written	
Department	3mE Department Maritime & Transport Technology	

OE44155	Mechanical and Hydraulic Excavation	5
Responsible Instructor	Dr.ir. R.L.J. Helmons	
Instructor	Prof.dr.ir. C. van Rhee	
Instructor	Dr.ir. G.H. Keetels	
Instructor	S.M.S. Alhaddad	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course entails the seabed excavation processes that are typically encountered in dredging and marine construction. The following content will be treated: - Introduction to dredging engineering and dredging equipment - intro to soil mechanics - Cutting mechanisms - Generic cutting model - Cutting of saturated sand - Cutter head spillage - Clay cutting - Rock cutting - Settling velocity - Subsea Rock Installation - Sedimentation & erosion - Jetting - Cable trenching	
Study Goals	By the end of this course, the students will be able to: - Explain the fundamental principles that govern rapid deformation of (submerged) soils - Apply the physical principles that govern excavation processes in dredging engineering - Evaluate the factors that affect the excavation process, including soil conditions and equipment used (mechanical, hydraulic or hybrid). - Assess the strengths and limitations of different models and assumptions used to simplify the analysis of the excavation processes.	
Education Method	Lectures	
Assessment	Written exam	
Department	3mE Department Maritime & Transport Technology	

OE44160	Limit States in ODE	2
Responsible Instructor	Dr. C.L. Walters	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The goal of this course is to introduce the definition and main analysis methods corresponding to principal limit states of offshore and dredging structures: fatigue, ultimate, accidental, and serviceability.	
Study Goals	-Identify relevant failure mechanisms of offshore & dredging engineering systems, based on demonstrated understanding of the working principle of such systems -Evaluate the fatigue and ultimate limit states of basic structural details of offshore & dredging engineering systems. -Familiarize with structural monitoring methods for assessment of limit states.	
Education Method	One lecture per week on average, some of which will be combined by workshops to improve (FE) modelling skills	
Assessment	There will be two main assignments, which will be graded as individual effort. The grade is 50% for each of the assignments. Ungraded assessments will also be given. Each report that achieves a grade between 5.0 and 5.74 can give corrections for one re-submission. The re-submitted grade may not be higher than 6.5.	
Department	3mE Department Maritime & Transport Technology	

OE44165	Sediment Transport in Dredging Engineering	5
Responsible Instructor	Dr.ir. G.H. Keetels	
Instructor	S.M.S. Alhaddad	
Contact Hours / Week x/x/x/x	HC 0.4.0.0 W 0.2.0.0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course addresses transport of sediment in dredging engineering.	
Study Goals	Explain the fundamental physical principles that govern sediment transport in dredging engineering. Apply the physical principles of sediment transport to practical problems in dredging engineering. Evaluate the factors that affect sediment transport in open and confined systems, including fluid properties, sediment properties, and boundary conditions. Assess the strengths and limitations of different models and assumptions to simplify the analysis of sediment transport in dredging engineering. Assess strategies for optimizing dredging operations	
Education Method	Lectures and formative assignment	
Assessment	Written exam	
Department	3mE Department Maritime & Transport Technology	

Year	2023/2024
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MSc ODE Ethics and Engineering (choose 1 out of 2 (3 ECTS))

WM0320TU	Ethics and Engineering	3
Module Manager	Dr. F. Santoni De Sio	
Instructor	M. Sand	
Co-responsible for assignments	M. Sand	
Contact Hours / Week x/x/x/x	4/0/4/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course is identical to the initial part of the course WM0329TU. You will explore the ethical and social aspects and problems related to technology and to your future work as professional or manager in the design, development, management or control of technology. You will be introduced to and make exercises with a range of relevant aspects and concepts, including professional codes, philosophical ethics, individual and collective responsibilities, value pluralism and relativism, responsibility within organisations, responsible conduct of companies and the role of law. You will analyse legal, political and organisational backgrounds to existing and emerging ethical and social problems of technology, and you will explore possibilities for resolving, diminishing or preventing these problems.	
Study Goals	After having completed the course you: - can better recognise and analyse ethical and social aspects and problems inherent in technology and in the work of professionals and managers active in the design, development, management and control of technology. - have insight into how these ethical and social aspects and problems are related to legal, political and organisational backgrounds. - are able to explore and assess possibilities for solving or diminishing existing and emerging ethical and social problems that attach to technology and the work of professionals and managers. - are better prepared to perform your future work as a professional or manager in the design, development, production and control of technology in an ethical and socially responsible way.	
Education Method	A series of 6 lectures and interactive work sessions with partial assessment, concluded with a written test.	
Literature and Study Materials	Chapters from the Reader Ethics and Engineering, available at Nextprint and as PDF files on Brightspace; articles available as PDF on Brightspace; Powerpoint slides, materials used in the working groups, and lecture notes.	
Assessment	Written exam (60%), presentation and active participation during the working group sessions (40%). In order to pass the course, a minimal grade of 5.5 should be obtained *both* in the exam *and* in the working group, and the average grade should be higher than 5.75. It is possible to re-take the exam without re-taking the working groups, in this case the working group grade will be kept. If you re-take the exam, it is also possible to re-attend the working groups to improve the final grade.	
Enrolment / Application	Enrolment via Brightspace is required for this course. This is needed in order to plan the workgroups. Please enroll as soon as possible and check Brightspace for more information on the enrolment on the enrolment procedure and timeline. The Brightspace page will be activated a couple of weeks before the start of the course.	
Remarks	The course is run twice each year in the first and third quarter. The course is identical to the initial part of the course wm0329tu (6 ects).	
Elective	Yes	
Category	MSc niveau	

WM0349WB	Philosophy of Engineering Science and Design	3
Module Manager	Dr. J.M. Duran	
Instructor	Dr. J.M. Duran	
Responsible for assignments	Dr. J.M. Duran	
Co-responsible for assignments	Ir. S.J. Zwart	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Course contents: (1) The goals of science; the character and scope of scientific claims. (2) The goals of engineering design; the nature of technical artefacts; the value-neutrality of technology. (3) The scientific method and the validation of scientific claims. (4) Methods of engineering design; the character and scope of design claims; the decision-making aspect of design. (5) The development of science; the objectivity of science; the notion of scientific progress. (6) The development of technology; social determinism and technical determinism. (7) The place and role of values in science and in engineering design.	
Study Goals	This course aims first of all to support students in developing a reflective and critical attitude with regard to empirical research underlying engineering science and engineering design at an academic level. Additionally, and in support of this primary goal, it aims to make students acquainted with views on the nature and goals of science and engineering design and how in these activities facts and values both have their role to play and how they interact.	
Education Method	The course is taught in the form of seven weekly sessions of a plenary lecture of two hours, and five seminars of two hours during which students articulate and discuss their answers to a number of questions in smaller groups. Each seminar will be prepared and chaired by a team of about three students. For this task general and specific instructions, recommendations and suggestions will be available.	
Assessment	Assessment is through (1) a final individual exam, (2) participation in the seminars and (3) performance in preparing and chairing a seminar session. The exam result forms 60 percent of the final grade, assessment of general seminar participation 20 percent and assessment of the preparing and chairing role also 20 percent. In the expectation that no COVID restrictions will be in place while the course is being offered, the exam will be a closed-book on-campus exam, which may be either hand-written or digital (Möbius). Which of the two will be announced at the start of the course.	
Tags	Artificial intelligence Ethics Philosophy Technology	
Category	MSc level	

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MSC ODE Specializations	
Introduction 1	Below are the specializations. Students place the emphasis on one specialization. However, the courses can be chosen throughout the specializations, in consultation with the master coordinator

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MSc ODE Dredging Engineering (7 ECTS)

Introduction 1

The dredging engineering specialisation involves the design of equipment for moving solids from, to, or over the bottom of the sea in water depths where offshore engineers normally work. The design of more conventional dredging equipment is discussed more or less along the way.

This specialization will always be included in any programme focused on deep sea dredging.

OE44050	CFD for Dredging & Offshore Engineering	3
Responsible Instructor	Dr.ir. G.H. Keetels	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Overall description: of the course Computational fluid dynamics (CFD) for dredging and offshore engineering. The target students are the dredging and offshore engineering master students who should have some understanding of CFD, should be able to program relatively simple flow solvers and judge CFD results of more complex flow problems in their field of application. The required background knowledge is very limited as these students do have a very limited number of lectures in numerical mathematics. The challenge is to explain and perform CFD as simple as possible with a focus on the physics and applications but with a minimum of formal mathematical derivations.	
Study Goals	after the course students should be able to 1. discretize a partial differential equations and boundary conditions with the finite volume method 2. explain the different features of elementary numerical schemes 3. program basic flow and transport solvers 4. motivate the choices for turbulence and multiphase modeling, numerical schemes, mesh features, boundary conditions in calculations of more complex flow problems 5. judge and evaluate computational fluid dynamics results. 6. present and write the skills achieved in the course short and clearly to other students that have attended the lectures.	
Education Method	Theoretical lectures will be combined with exercises and discussions during the classes. A matlab programming homework exercise will be provided just after the beginning of the course. Students will select a problem they like to work on and should extend an elementary CFD code in duo teams. PhD researches in the dredging and offshore group can assist with the homework exercises if necessary. Some CFD case studies of more advanced flow problems will be presented during the classes where students will be asked to discuss with each other how to tackle the problem.	
Books	Numerical computation of internal & external flows, Charles Hirsch	
Assessment	Written (weight 0.5) and Nutshell paper about the code extension max 4xA4 (weight 0.5)	
Remarks	Old course code: OE4628	
Department	3mE Department Maritime & Transport Technology	

OE44110	Subsea Engineering	4
Responsible Instructor	Dr.ir. R.L.J. Helmons	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	Course is designed for students with a background in one or more of the following programmes: - Offshore and Dredging Engineering - Maritime Technology - Mining Engineering In general this means that a bachelor background in Mechanical, Maritime, Civil or Mining Engineering would suffice. Other backgrounds are possible, but it is advised to contact the teacher.	
Course Contents	The ocean floor contains vast mineral resources that may be used in the development of renewable energy technologies. These minerals are expected to be in high demand due to the de-carbonization of the global economy and the transition to a net-zero economy. While the blue minerals may hold the key to a greener future, mining the deep seas raises many questions and dilemmas. Can these minerals be mined without causing serious harm to the unique ecology of the deep oceans? Who will be allowed to exploit these minerals, and on what conditions? What rules and incentives are needed to attract responsible investors that can pay for the necessary scientific and technological development? Within this course, we will provide a holistic view on deep-seabed mining. Students will work on the challenges associated with marine minerals. The following aspects will be discussed: - Introduction and historical overview of wet mining - Exploration for (deep)sea minerals - Excavation of minerals - Vertical transport and riser systems - Environmental impact, e.g. turbidity plumes, underwater noise, seabed alteration - Monitoring The state-of-the-art will be discussed and students are encouraged go beyond the state of the art on several of the challenges associated with marine minerals: - Environmental monitoring and operation supervision - Exploration technologies for mineral identification and resource estimation - Production concepts and technologies - Mineral processing and value chain possibilities	
Study Goals	- Students are able to explain terminology related to deep-sea mining - Students are able to motivate design decisions for deep-sea mining components, with an emphasis on the economic production, reliability and environmental impact - Students are able to motivate design choices for complete deep-sea mining system and peripheral systems, with an emphasis on operability, reliability and environmental impact	
Education Method	Classroom lectures and excersises. Various guest lectures of experts will be included to provide a balanced view on the many aspects of deep-seabed mining, e.g. marine biology, marine geology, finance, mineral processing	
Literature and Study Materials	Lecture notes and other relevant publications will be made available on Brightspace. Subsea Engineering Handbook, authored by Bai. Available as e-book through DUT library (free access). Recommended other materials: Laptop for exercises (e.g. Excel, Python, Matlab)	
Assessment	Group assignments (2-3 students) regarding design concepts of deep-seabed mining systems for various types of mineral deposits. If the assignment is satisfactory, an individual oral defense can be scheduled. Both the assignment and the individual oral exam need to be satisfactory in order to pass the course. Final grade is based on 50% deep-sea mining assignment and 50% individual oral defense.	
Department	3mE Department Maritime & Transport Technology	
Judgement	Grades are assigned based on the results of a report on the field development assignments and an oral defense of the assignments/report. All components need to be satisfactory in order to pass the course.	
Contact	Dr. ir. Rudy Helmons Faculty of Mechanical, Maritime and Material Engineering (3mE) Department of Marine and Transport Technology (M&TT) Building 34 Room B2-290 +31 (0)15 27 84746 R.L.J.Helmons@tudelft.nl	

Year

Organization

Education

2023/2024

Mechanical, Maritime and Materials Engineering

Master Offshore and Dredging Engineering

MSc ODE Structural Analysis & Design (15 ECTS)	
Introduction 1	edit text

MT44010	Non-metallic materials in Marine Structures	5
Responsible Instructor	L. Pahlavan	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course you will learn about fiber-reinforced composite (FRC) materials and structures. Applications of FRCs are rapidly growing in the maritime sector due to their light-weight and superior corrosion resistance compared to their metallic counterparts. For those who have interest in structural design and analysis, it is important to understand how these materials can be designed, manufactured and recycled, applied to modern maritime of offshore structures, and inspected/monitored. MT44010 includes two parts: (i) fundamentals of FRC materials; and (ii) analysis and assessment of structures made of FRCs. The first part of the course introduces the manufacturing processes, recycling methods, microstructural characteristics, and mechanical behavior of composites highlighting the process-microstructure-property relationship. The second part focuses on the analysis of structures made of FRCs, and fundamentals of the conventional and advanced measurement methods for assessment of composites. The course contains two project assignments, and an experimental demonstration session.	
Study Goals	Upon completion of the course, students should be able to: 1. Describe manufacturing and recycling processes for composite materials 2. Relate microstructure and properties of composite materials 3. Estimate strength of composite lamina from both micro- and macro-mechanics theories 4. Analyze elastic behavior of composite lamina and laminates using Classical Laminate Theory 5. Assess bending, vibration, and buckling behavior of structures made of composite material 6. Select suitable measurement techniques for assessment and monitoring of FRCs	
Education Method	Lectures, project assignments, and self-study	
Literature and Study Materials	Daniel and Ishai (2006), Engineering Mechanics of Composite Materials, Oxford university press Reddy, J. N. (1997), Mechanics of Laminated Composite Plates. CRC Kassapoglou, C. (2013), Design and Analysis of Composite Structures, Wiley	
Assessment	Project assignments (80%) Presentation of project results and oral assessment (20%)	
Department	3mE Department Maritime & Transport Technology	

MT44085	Ultimate Strength of Marine Structures	5
Responsible Instructor	Dr. C.L. Walters	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The goal of this course is to assess the safety of a marine structure subject to the ultimate limit state. A solid understanding of the ultimate limit state and ways to model it are developed. The main focus is on buckling and fracture. Analytical and numerical methods will be applied to both failure mechanisms.	
Study Goals	- explain which load and structural parameters influence the buckling and ultimate strength and the reason why - calculate the buckling and ultimate strength of a stiffened panel using hand calculations and the finite element method - judge modelling choices in non-linear finite element methods applied to stiffened panels - assess the safety of the grillage against brittle fracture - assess the safety of the grillage due to an accidental overload	
Education Method	Two lectures a week, some of which will be combined by workshops to improve (FE) modelling skills.	
Assessment	Two reports (made in groups of 2 persons) and individual oral review. All weigh equal.	
Remarks	Old course code: MTM815-15	
Department	3mE Department Maritime & Transport Technology	

OE44085	Fatigue Strength of Marine Structures	5
Responsible Instructor	Dr.ir. J.H. den Besten	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	Marine structures active in inland, coastal, offshore and deep(sea) waters are exposed to cyclic mechanical loading, forces and moments, both environment (wind, waves) and service (operations, machinery) induced. The response in terms of stress, strain or energy is cyclic by nature accordingly, meaning fatigue, a cyclic loading & response induced local, progressive, structural damage mechanism is a governing limit state. To be able to assess the marine structure integrity & longevity in this matter, course contents include: Fatigue fundamentals, i.e. physics and damage process phenomena in: materials at defect locations (stress concentrations at micro- and meso-scale); different sources and different stages in the fatigue & fracture process (crack initiation, growth and propagation, fracture). structures at notch locations (stress concentrations at macro-scale) as part of structural members or at the member connections; particular attention will be paid to arc-welded joints in that respect, since marine structures are predominantly assemblies consisting of plates, shells and tubular members. Macroscopic (notch) stress concentrations, distributions and intensities: as part of structural members. at (welded) structural member connections, consisting of a cyclic mechanical- and quasi-constant thermal residual part. Fatigue resistance: materials (plane geometries) structures (notched geometries: welded joints) at small-, large- and full-scale testing (measurement techniques) damage criterion life time relations in different ranges (low-, medium- and high-cycle fatigue); formulations and parameter estimates using regression analysis (least squares- and likelihood approaches). micro- and macro (short and long) crack growth relations. Fatigue assessment concepts: damage criteria using an intact geometry parameter (nominal stress-, structural hot spot stress- and effective notch stress/ strain concept). damage criteria using a cracked geometry parameter (crack tip stress intensity- and equivalent stress concept). Particular attention will be paid to analytical formulations and numerical approaches; concept capabilities and limitations. Fatigue influence factors: material and geometry (strength, surface integrity, volume, welding improvement techniques), environment (temperature, corrosion), loading & response (min/max ratio, mean stress, amplitude variability (time domain analysis, cycle counting, spectral approaches, scatter diagrams, cumulative damage hypotheses), multi-axiality including proportionality (in-phase, out-of-phase). Fatigue reliability for (risk based) inspection and maintenance purposes: time dependent limit state functions, (joint) design variable statistics, level 1, 2 and 3 analysis, 1st and 2nd order analytical reliability methods and Monte Carlo simulation. Structure integrity & longevity: fatigue strength calculation and life time estimation for assessment and design using different (design) philosophies, concepts, rules, standards and guidance.	
Study Goals	Marine structures active in inland, coastal, offshore and deep(sea) waters are exposed to cyclic mechanical loading, forces and moments, both environment (wind, waves) and service (operations, machinery) induced. The response in terms of stress, strain or energy is cyclic by nature accordingly, meaning fatigue, a cyclic loading & response induced local, progressive, structural damage mechanism is a governing limit state. To be able to assess the marine structure integrity & longevity in this matter, course objectives include: Theory (fundamentals): to identify and describe the fatigue physics and damage process phenomena in materials & structures to establish and calculate the macroscopic fatigue characteristic structural response parameters to investigate fatigue resistance testing considerations and characterise damage criterion life time relations to model fatigue physics mathematically based on different damage criteria developed in the course of time to assess fatigue resistance consequences of relevant influence factors and incorporate models in fatigue damage criteria to examine 1st and 2nd order as well as Monte Carlo simulation based reliability methods at level 1, 2 and 3 for (risk based) inspection and maintenance purposes Validation (experiments): to evaluate dedicated fatigue testing techniques to analyse and process measurement data and establish fatigue resistance parameters Application (engineering): to identify and classify fatigue sensitive locations in structures to select an appropriate fatigue assessment concept to establish and improve marine structure integrity & longevity, i.e. to calculate the fatigue strength and corresponding life time estimate for analysis and design purposes	
Education Method	Lectures (blended)	
Assessment	For qualification purposes assignments will be provided. Analysis, design and optimisation tasks to be performed require use of MATLAB and ANSYS. The assignment reports will be assessed and graded. For evaluation purposes, an appointment will be scheduled in order to discuss the results and to verify up to what extent the course objectives are achieved.	
Tags	Analysis Modelling Structural Mechanics	
Department	3mE Department Maritime & Transport Technology	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Offshore and Dredging Engineering

MSc ODE Offshore Renewable Energy (12 ECTS)	
Introduction 1	The offshore renewable energy specialisation addresses the required knowledge and its effective use needed by the offshore industry to accomplish the technical challenges of the emerging offshore renewable energy sector. These technical challenges have a direct consequence in the shift towards a new offshore industry powered by renewable energy as part of the global energy transition.

OE44097	Offshore Support Structures	6
Responsible Instructor	Ir. J.S. Hoving	
Instructor	Dr.ir. P. van der Male	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The primary objective of this course is to enable students to produce conceptual engineering designs for offshore support structures, to be used for offshore wind and other offshore energy developments, accounting for ultimate strength and fatigue criteria. During this course, students gain a thorough theoretical understanding of the design process, its structuring, available design methodologies and the philosophy behind the design codes. Furthermore, this module addresses offshore, geotechnical, structural and reliability aspects in coherence to analyse the behaviour of offshore support structures. Throughout the course students work on a group design project, where each student team designs an offshore support structure from scratch up to a preliminary design by performing a full offshore support structure design-cycle. This design-cycle includes I) the assembly of the structural configuration, II) the determination and assessment of the permanent, variable and environmental loads on the structure, III) the design and analysis of its foundation and IV) the static and dynamic structural analysis of the structure based on ultimate strength and fatigue principles. By means of this group design project, students will learn to judge their design against ULS and FLS design criteria and be able to modify and improve their design based on static and dynamic structural analyses. Thereby, students will be able to analyse and evaluate the dimensioning and performance of structural components for strength and stability. Furthermore, students learn to predict how changing (parts of) their design will influence the corresponding loads and based on their insight in these loads, be able to defend their design choices. Moreover, students will learn what the practical implications of their design choices are for operations and other feasibility and life-cycle aspects. To reinforce the understanding of these aspects, the course program also includes a visit to an offshore structures fabrication yard. During this module, students will not only learn how to design an offshore structure, but they also learn how to collaborate effectively in a team and will further develop skills and competences to motivate their assumptions and provide argumentation of their design choices by accounting for relevant moral values, norms and principles.	
Study Goals	After following this course on Offshore Support Structures (OSS), students are able to: > Perform a full design-cycle as a part of the OSS design process > Assemble the structural configuration of offshore support structures > Assess the permanent, variable and environmental loads on OSS > Design and analyse (piled) foundations for OSS > Analyse and evaluate the structural design of OSS based on ULS and FLS analysis > Perform a natural frequency analysis and a frequency response assessment for OSS > Evaluate the influence of life-cycle aspects on the design of OSS, including fabrication, transport, installation, operations management & decommissioning Critical thinking, analytical thinking and problem solving are the main academic skills addressed in this course. Furthermore, students will collaborate intensively with their colleagues in a module-wide group design project thereby developing organizational and interpersonal skills. In addition, students develop written communication by presenting and reflecting on their designs in a compact design report, requiring to focus on the essentials of their design and their work.	
Education Method	The education in this course includes problem-based lectures, workshops, a group design project and self-study. In total, this course has 8 contact hours per week, consisting of: - two 2-hour lectures (2x45 min) focused on comprehension and application of theory and methodology of offshore support structures design, and - a 4-hour workshop (4x45 mins) focused on problem analysis and evaluation, treating and discussing example problems, as well as to provide feedback and reflection on the group design project as well as allowing for discussion and Q&A regarding the group design project. The group design project is performed in groups and consists of 3 phases. Each phase comes with a formative assessment & feedback session per group in interaction with the lecturers and/or your peers, followed by a final deadline at the end of the teaching period at which the assignment is graded. The groups generally consist of 3 or 4 students, but the group size may be adapted depending on cohort size. Additionally, under reservation of unforeseen and/or external circumstances, the module includes an excursion to a fabrication yard where offshore support structures are constructed. Here note that the number of participants to this excursion is generally limited by the capacity at the yard and the excursion is therefore not a compulsory part of the module.	
Course Relations	This course is part of the MSc Offshore & Dredging Engineering (ODE) and the Offshore track of the European Wind Energy Master (EWEM), but can also be followed as an elective in any MSc related to structural and/or offshore engineering. This course runs parallel and in sync with the unit Design of Offshore Structures, embedded in the module CIEM4120 Offshore Structures, which is part of the track Hydraulic and Offshore Structures (HOS) of the MSc Civil Engineering. This course requires no particular compulsory pre-knowledge, however the course OE44150 Marine Hydromechanics provides useful information for this course.	
Literature and Study Materials	The lecture material for this course consists of the lecture slides, lecture notes, papers and chapters from relevant literature, a basis for design document for the group design project and the stick-model hand-out, which will all be available through Brightspace. Additional reading material that provides useful information in support of the course are: > The books Handbook of Bottom Founded Offshore Structures, parts 1 and 2, by prof.dr.ir. Jan H. Vugts, available online through the website of the Publisher Eburon. > The book Hollow Sections in Structural Applications, by prof.dr.ir. J. Wardenier, available at VSSD.	
Assessment	Participants of this module are summatively assessed through a Group Design Project, worth 4 credits, and an individual written exam, worth 4 credits. The final grade for the course is the weighted average of the Group Design Project (2/3) and the individual written exam (1/3). To pass the course, participants must score a pass grade, i.e. at least a 6.0 (rounded to halves), on both the project and the exam. Group Design Project: The final deadline for the Group Design Project is set at the end of the Q3 education period. If you fail to get a pass grade for the assignment, but the assignment grade is equal to or higher than a 5.0, you are allowed to improve your grade by submitting an addendum to your assignment report in the following education period, i.e. during Q4.	

	Here, please note that the maximum grade that can be obtained after submitting an addendum to improve an insufficient grade is 6.0. Written exam: The regular individual written exam for this course is organised during the exam period of Q3, and a re-exam is organised in the exam period of Q4. The written exam is an open-book exam, meaning that any written or printed material is permitted during the exam sessions. If due to any unforeseen circumstances the written exam for this course may not be given in a plenary on-campus session, alternatively the written exam for this course may be given as an online open-book exam through Brightspace. To participate in the online exam you must be enrolled for this course in Brightspace. For further practicalities regarding any online examinations, you will be informed through Brightspace.
Permitted Materials during Tests	The written exam is an open-book exam, meaning that any books, slide print-outs, personal notes, i.e. any written or printed material is permitted during the exam sessions. However, to limit the amount of paper waste, a maximum of 10 A4-pages (written or printed) are allowed during the exam. This rule does not include books. Additionally, it is allowed and advised to bring a scientific calculator to the exam. Nevertheless, any devices that allow for communication are not allowed, i.e. no laptops, pads or phones.
Enrolment / Application	For this course, enrollment in Brightspace is required; all information of the course and communication regarding this course is provided through Brightspace. If due to any unforeseen circumstances a lecture or workshop for this course may not be given in a plenary on-campus session, these on-campus lectures may be replaced by online lectures. For further practicalities regarding any online lectures, you will be informed through Brightspace.
Remarks	Through the past years, this course has seen several updates and improvements, and thereby updated course codes. Previous course codes are: OE4651, OE4651-12, OE4651-15, OE44095 and OE44096

OE44097 Toets 1	<i>Exam Offshore Support Structures</i>	2
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OE44097 Toets 2	<i>Assignment Offshore Support Structures</i>	4
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OE44100	Floating Structures & Offshore Moorings	6
Responsible Instructor	Dr.ing. S. Schreier	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	MT44020 Motions & Loadings of Structures in Waves Ordinary differential equations (numerical solution of initial value problems) Dynamics of 1 and 2 DoF mass-spring-damper systems	
Course Contents	A generic (and fundamental) approach for the analysis of the dynamics of offshore floating concepts and their mooring systems is presented and discussed together with the required engineering tools. Typical examples that will be discussed during the lectures are the dynamics of structures for the oil&gas industry as well as other floating structures, e.g. for marine renewable energies, the associated workability in the dynamic environment, and the deterministic design of a mooring system for a floating system including the verification/validation through model tests. Based on equations of motion and the stochastic character of the waves the response of these complex systems is analysed and interpreted. In this context, the effects of modelling choices on accuracy and system performance are addressed and related to engineering practice. Different analysis techniques are discussed and practised during the workshops and through assignments. The following topics are addressed: Floating structures: - Dynamics of typical offshore structures used in the oil&gas industry as well as in offshore renewable applications - Derivation of equations of motions for such structures - Structured approach in the analysis and importance of modelling choices - Consequence of stochastic nature of waves for floater response - Identification of limiting response and limiting sea states (workability) Mooring systems: - Typical mooring systems and mooring line configurations - Design criteria and safety factors for mooring design - Interpretation and relevant combination of environmental conditions (metocean data input) - Ultimate strength and fatigue issues of mooring systems - Current industry standards and design principles	
Study Goals	After successfully completing this course, you are able to: For a floating multibody system in waves 1. Create a dynamic system model 2. Analyse the system characteristics and response in waves 3. Evaluate the workability of the system based on metocean data 4. Evaluate the influence of modelling choices on the model performance For offshore mooring systems 5. Analyse mechanical properties of (basic) mooring configurations 6. Evaluate mooring system designs for different applications	
Education Method	This course uses a problem-based approach and is based on lectures, workshops, and mandatory assignments. During the lecture period the assignments are handed out and completed in groups. The workshops provide time for assisted work on the assignments. If still necessary, the course will be given online with emphasis on students reading lecture material and working on assignments as well as online Q&A sessions.	
Literature and Study Materials	Floating Structures - Journée, JMJ, Massie, WW, Huijsmans, RHM (2015). Offshore Hydromechanics, 3rd edition. available on the Brightspace page of this course (relevant chapters to be announced during the course) Offshore Moorings - Ma, K-T, Luo, Y, Kwan, T, Wu, Y (2019). Mooring System Engineering for Offshore Structures. available via Science Direct: https://www.sciencedirect.com/book/9780128185513/mooring-system-engineering-for-offshore-structures (relevant chapters to be announced during the course) Information on Ocean Waves (not mandatory) - Holthuijsen, LH (2007). Waves in Oceanic and Coastal Waters. e-book available via TU Delft Library: https://tudelft.on.worldcat.org/search?queryString=%22Waves+in+Oceanic+and+Coastal+Waters%22#/oclc/663973262	
Assessment	Assessment consists of two parts. Part 1 are the assignments given during the lecture period. There are 4 assignments in total. All 4 assignments are mandatory. The first 2 assignments will not be graded. The solution of the first 2 assignments will be published to provide feedback. The last 2 assignments are graded separately. The total grade for the assignments is given by the weighted average of the separate grades for the 2 graded assignments. Part 2 is the written exam at the end of the lecture period. The final grade of the course is the weighted average of the combined grade of the assignments and the written exam. Only the final grade is registered in Osiris. To successfully complete this course the grade per part, i.e. assignments and exam, has to be at least a 5.0. The grades obtained per part remain only valid within the same academic year. In case of a resit in a later academic year, the whole course has to be retaken comprising both assignments and final exam. The written exam will be as follows: Closed-book written exam on campus. In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be: Open-book online written exam. For details on permitted material in either case see item "Permitted Materials during Tests".	

For further rules and regulations we refer to the relevant website
(<http://studenten.tudelft.nl/informatie/faculteitspecifiek/3me/onderwijsinformatie-3me/organisatie/regelingen-reglementen/>).

Permitted Materials during Tests

In case of on-campus written exam (closed book)
- permitted material: calculators (but no graphical calculator)
- not allowed: books, lecture material, notes

In case of online written exam (open book)
- permitted material: all lecture materials, books, notes, internet
- not allowed: any communication with other persons including, but not limited to, fellow students (questions to the lecturer are allowed)

Department

3mE Department Maritime & Transport Technology

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Offshore and Dredging Engineering

MSc ODE elective courses 1st year (courses from other specializations can be chosen as elective)

AE4W31	Floating Offshore Wind Energy	3
Responsible Instructor	Dr.ir. A.C. Viré	
Instructor	Dr.ir. D.A.M. De Tavernier	
Contact Hours / Week x/x/x/x	0/0/0/2	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course introduces the topic of floating offshore wind energy, a fast-growing field in offshore wind energy, which consists in placing offshore wind turbines on a floating support structure. The course highlights the need for floating offshore wind, the main challenges compared to conventional bottom-fixed offshore turbines, and the impact on the industry as a whole. More specifically, topics covered in the course are: - Why floating offshore wind, ongoing projects and trends - Aerodynamics and control of floating offshore wind turbines - Hydrodynamics of floating offshore wind turbines - Small-scale and full-scale testing - Logistics and project development of large floating offshore wind farms - Alternative wind turbine concepts for floating foundations	
Study Goals	After completing the course, the students will be able to: - Explain the relevance and need for floating offshore wind energy in the current energy transition - Derive appropriate aerodynamic and hydrodynamic models for floating offshore wind turbines - Calculate the effect of a floating motion on the turbine performance - Define guidelines for testing floating offshore wind turbines in wind tunnels and wave tanks - Explain the industrial challenges in developing a large floating offshore wind farm - Explain the differences between bottom-fixed and floating turbines in the overall supply chain - Explain pros and cons of different wind energy concepts in a floating environment	
Education Method	The course consists of live lectures, assignments and self-study materials on Brightspace. Guest lecturers from industry and other Faculties/Universities will also participate in the course to reflect the multi-disciplinary nature of floating offshore wind energy.	
Assessment	The course is assessed through a written exam at the end of the course and the assignments performed during the course.	

MT44001	Mechatronics in MT	5
Responsible Instructor	A. Coraddu	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	BSc courses on control engineering (incl. Proportional-Integral-Derivative control), hydromechanics, marine engineering, and signal analysis and processing. Software Skills: Matlab/Simulink.	
Course Contents	Overview: Unlock the fascinating intersection of mechanics, electronics, and information technology in maritime settings through this cutting-edge course. Our main objective is to guide you in designing a Dynamic Positioning (DP) system a complex mechanism to stabilize a model boat against various natural disturbances like waves, wind, and currents. What You'll Learn: Integrated Systems Approach: Explore the synergy between mechanics, electronics, and IT in maritime projects. Dynamic Positioning (DP) System: Delve into designing and constructing a DP system to stabilize a model boat against environmental variables. Hands-on Experience: Use the Matlab/Simulink environment to create your DP model. Real-world Implementation: Transition from theoretical design to hands-on testing in our state-of-the-art lab. Critical Evaluation: Assess the strengths and weaknesses of your DP model in terms of performance and reliability. Core Components: Marine Engineering: Understand vessel design, propulsion, and other marine systems fundamentals. Control Engineering: Gain in-depth knowledge of control theory, including PID controls essential for dynamic systems. Digital Signal Processing: Master techniques for analyzing, interpreting, and manipulating signals critical for effective control systems. Why Take This Course? Interdisciplinary Learning: Equip yourself with a unique blend of skills from mechanics to digital signal processing, setting a robust foundation for your engineering career. Hands-on Application: Bridge the gap between theoretical knowledge and real-world applications through lab experiments. Career Relevance: Prepare for an ever-evolving marine, aerospace, and control systems engineering job market. Holistic Understanding: Gain a 360-degree perspective on maritime mechatronic systems, with the DP system serving as a quintessential example. Performance Analysis: Acquire the skills to critically evaluate and optimize complex engineering systems for real-world challenges. By the end of this course, you'll have a working DP model and a comprehensive understanding of how to design, analyze, and evaluate complex maritime mechatronic systems.	
Study Goals	1 - Develop an optimal DP solution for a given ship by understanding the theory and practice. 2 - Apply the basics of computer-based modeling, measuring, and control systems. 3 - Apply basics of digital signal processing relevant to mechatronics systems (sampling, filtering, signal resolution, A/D and D/A conversion). 4 - Describe the basic concepts of mechatronics and the working principles of sensors and actuators. 5 - Analyse and interpret the performance of mechatronics systems. 6 - Evaluate the differences between the theory and practice of control systems. 7 - Evaluate the strengths and weaknesses of mechatronic systems.	
Study Goals continuation	After successfully completing the course, students will be able to: - Describe the basic principles of mechatronics - Describe sensors and actuators in general and the ones specific to the project, - Apply basics of signal processing relevant to mechatronic systems (sampling, filtering, signal resolution, A/D and D/A conversion), - Use and tune computer-based measuring and control systems, - Analyse and interpret the performance of a miniature Dynamic Positioning system regarding theoretical and practical aspects, - Evaluate the strengths and weaknesses of a mechatronic system that the students set up.	
Education Method	Lectures + Assignments. In the lectures, course material from the course books will be elaborated upon using lecture sheets and interactive Q&A sessions. The assignments enable students to obtain experience with the required calculations during the course period. In this way, the assignments support students in developing the required insight into the course contents to submit the deliverables.	
Books	Bolton, W. (2003). Mechatronics: electronic control systems in mechanical and electrical engineering. Pearson Education. Karnopp, D. C., Margolis, D. L., & Rosenberg, R. C. (2012). System dynamics: modeling, simulation, and control of mechatronic systems. John Wiley & Sons. Cetinkunt, S. (2015). Mechatronics with experiments. John Wiley & Sons. Fossen, T. I. (1994). Guidance and control of ocean vehicles. John Wiley & Sons Inc. Wadoo, S., & Kachroo, P. (2016). Autonomous underwater vehicles: modeling, control design and simulation. CRC Press. Karris, S. T. (2006). Introduction to Simulink with engineering applications. Orchard Publications.	
Reader	Reader (available on Brightspace). Week assignments (available on Brightspace). Additional material supplied on Brightspace.	
Assessment	Students are evaluated based on their performance with respect to the following deliverables: D1: Demonstrate a selected DP solution using a working (online) control system computer model - 30%. D2: Demonstrate multiple DP solutions using a working computer model (offline) - 30%. D3: Document your solutions in a report, discuss the validity of the offline model as a representation of the online system, and reflect on the learning process - 40%.	
Remarks	The maximum capacity of MT44001 is 35 students per period. The reason is the availability of the flume tanks, boats, required charging time, etc.	
Department	3mE Department Maritime & Transport Technology	
Contact	Dr Andrea Coraddu - a.coraddu@tudelft.nl	

OE44090	Introduction to Computational Dynamics of Offshore Structures	4
Responsible Instructor	J.O. Colomes Gene	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	CIE4140 Structural Dynamics	
Course Contents	In this second year you will learn about solving dynamical system numerically. Each week there is a ~2-hour theory lecture and a ~2-hour practical session. The theory lectures will be given as a set of videos with the supporting slides. There will be in-class interactive tutorial lectures where some examples will be solved and you will have the opportunity to ask questions and discuss with the group. Discussions are also encouraged through the Brightspace discussion forum. The topics that will be covered during the lectures are: - How ODE solvers work - Understanding the different structural elements - Systems with constraints - Finite difference in 1-dimensional - Discretization of 1D structural models; rods & beams - Modeling loading on structure, focusing on spectral loads like waves - Geometrically nonlinear models in 1D - System with frequency-dependent parameters (added mass and damping) During the practicals you will work on the assignments in small groups. In total there will be two assignments, costing about 15-20 hours each (while this might seem like a lot of time, I believe that the best / only way to learn something is to do it yourself). Each assignment puts the theory explained during the lecture into practice by applying it to solve a dynamical problem. While default problems will be provided each week, you are free to choose your own problem to solve as long as it requires certain modeling technique. Examples of problems are: - a ship with a crane - a wind turbine - a mooring system - a bridge Overall, this course will strengthen your understanding of dynamics, improve your programming skills and will give you a good understanding of the basics of the numerical aspects of related to solving dynamics system numerically. This knowledge will help you greatly during your Master Thesis and, afterwards, while working.	
Study Goals	At the end of this course you will be able to: - LO1: Construct a model that represents an offshore engineering application, limited to models that can be constructed from a combination of: point masses, rigid bodies, rods, Euler-Bernoulli beams or geometrically non-linear rods. - LO2: Derive the equations of motion of the model, subject to typical offshore loads such as: wind, waves and currents. - LO3: Define different numerical methods for ODEs and PDEs, including: Finite Differences and Finite Element methods. - LO4: Implement the numerical methods and solve the problem using Matlab, Python or other programming languages. - LO5: Analyse the results by: validating against analytical solutions, identifying the range of applicability of a given method, evaluating errors and assessing the convergence of the solution.	
Education Method	See above	
Assessment	The method of assessment will be: - 50% of the score based on the averaged score of one assignments - 50% of the score based on written exam on campus If measures resulting from COVID-19 need to be implemented, the assessment will be as follows: - 50% of the score based on the averaged score of one assignments - 50% of the score based on a remote open book written exam The average score of all assignments and the written exam score need to be higher than a 6 to pass the course. Students will have the opportunity to take a resit exam at the end of the following quarter. Students can hand-in an individual addition to the assignments that will be rewarded with a maximum grade of 6.0. Deadline for the addition is 2 weeks after the announcement of the grades. The exact instruction for the addition will be announced on Brightspace.	
Department	3mE Department Maritime & Transport Technology	

OE44105	Marine Pipelines	4
Responsible Instructor	Dr. K. Anupam	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Marine Pipelines concentrates on different aspects of subsea pipeline design: Pipeline Design: The internal and structural design of pipelines for oil, gas and multi-phase (liquid gas) flows. Pipelines are dimensioned based upon flow in relation to properties of the transported material as well as capital expenditure and operating costs involved. The need for and means of providing thermal insulation is discussed including the measures of assuring flow in pipelines. Pipeline Route Selection: Routing of pipelines through the sea as well as their shore approaches are covered. Special attention is given to sub-sea tie-ins, pipeline and cable crossings, pipeline protection from fishing gear, shore approaches and pipeline trenching. The consequences of pipeline temperature changes and upheaval buckling are integral aspects of this topic as well as the on-bottom stability (pipelines on or in the sea bed). Pipelines Installation / Construction: This segment presents current and new technologies for the installation of pipelines in varying water depths, ranging from a few meters to depths measured in kilometers coupled with the role which installation plays in the design of a pipeline. Special attention is given to supporting finite element analysis (FEM) calculations, construction start-up, sea-bed lay-down, tie-ins and to welding technology. A classroom exercise is included as an integral aspect of the knowledge gained.	
Study Goals	Participants completing this course successfully will be able to function at an advanced to superior knowledge level productively and quantitatively in marine pipeline design teams.	
Education Method	Classroom/Online lectures.	
Assessment	Written exam	
Remarks	Old course code: OE4653	
Department	3mE Department Maritime & Transport Technology	
Contact	Kumar Anupam	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Offshore and Dredging Engineering

MSc ODE 2nd year

Introduction 1

The second master year consists of:

Period 1: specialisation courses and/or electives and/or JIP

Period 2: Literature Research

Period 3 en 4: Thesis

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Offshore and Dredging Engineering

ODE Graduation Project (Obligatory 45 ECTS)

OE54010-22	ODE Literature Research	10
Responsible Instructor	Dr.ir. P. van der Male	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Course Contents	This course serves as preparation for your master graduation project (OE54035). You are expected to do a review of the available literature for your intended topic of graduation, which result in the framework for your research. The report should consist of the following components: - A background description, or a description of the field. - A general problem statement. Why further research for this field is needed? - An overview of the available relevant literature. What research have been done? What problems have been considered? What are the recent developments? Other sources you could use are interviews, discussions with an expert panel, simple benchmark calculations etc. - A specific research objective, research question or hypothesis. What should be the outcome of your research? - A description of your research approach. How do you intend to reach your objective? A summary of your literature review can serve as the introduction of your graduation report. The procedure to find a topic for your literature review and graduation is as follows: - Contact the professor, or other academic staff members, of your specialization to see if there is a project available that you might like. TU Delft is frequently contacted by problem owners for such projects, but this is not guaranteed. In a discussion you can both find out what subject interests you and if there is a project available that relates to that subject. In most cases there is a rough problem description available of about ½ an A4. If not, most academic staff members can tell you where to search or who to contact to help find a suitable project. - Another option is to contact a problem owner (company/PhD student/Staff Member) yourself (e.g. through your own network). If they agree to take you as a graduation student on a subject you find interesting, you should write a short description of about ½ an A4 to discuss with the specialization prof. This is usually sufficient to check if a problem is a viable subject to graduate on. - Based on the initial description and discussions with all involved parties, you should extend the description to a more detailed Plan of Approach that ensures that both you and your supervisory team agree on what you will do in the next 9 months. If all parties agree with your Plan of Approach the initial phase is concluded and it is time to set the date for a kick-off meeting. Usually at this point, you should also be assigned a daily supervisor (the problem owner) and an academic supervisor (TUDelft). It is advised to start about 3 months in advance with these preparations.	
Study Goals	By the end of this course, you should be able to: - Search the literature systematically; - Formulate a research question; - Select representative scientific sources from several (for example, economic, ethical, environmental, medical, social, health) perspectives relevant to the assignment; - Draw conclusions related to the research problem and give recommendations towards new research opportunities; - Write a literature review in academic English; - Demonstrate a scientific attitude; - Manage your own learning process (including time management).	
Education Method	Self study	
Assessment	Literature Research (written report).	
	The assessment is based on the Rubric MSc Literature Research available in the 3mE student portal.	
Department	3mE Department Maritime & Transport Technology	

OE54035	ODE Thesis	35
Responsible Instructor	Dr.ir. P. van der Male	
Contact Hours / Week x/x/x/x	0/x/x/x	
Course Contents	If a student undertakes the graduation project at a company, a graduation agreement should be signed. In this respect, the template of the TUD-3mE graduation agreement should be used, which is available at the following link: https://www.tudelft.nl/en/student/faculties/3me-student-portal/education/related/student-forms/msc-forms/.	

Year	2023/2024
Organization	Mechanical, Maritime and Materials Engineering
Education	Master Offshore and Dredging Engineering

MSc ODE elective courses 2nd year (min. 15 ECTS)

CIEM4302	Cold Regions Engineering	5
Responsible Instructor	Ir. J.S. Hoving	
Responsible Instructor	Dr.ir. H. Hendrikse	
Instructor	Dr.ir. H. Hendrikse	
Instructor	Ir. J.S. Hoving	
Contact Hours / Week x/x/x/x	4 (lectures) + 2 (workshop / question hour related to assignment)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Cold regions engineering relates to all engineering activities related to coping with low temperatures, specifically freezing temperatures, in the cold regions of our planet. The formal education of most civil engineers does not include an adequate understanding of the unique characteristics that work in these cold regions brings. Challenges include engineering problems related to soil mechanics, hydraulics, heat flow, materials and construction, among others, in cold environments. Therefore, this module addresses a broad range of cold regions engineering design and construction activities. Topics covered in this module are: - Ice features and ice regimes; - Ice morphology, ice physics and ice mechanics; - Thermodynamics of ice growth and icing; - Cold regions structures and development; - Ice loads on port- and offshore structures; - Dynamic ice-structure interaction; - Material behavior at low temperatures*; - Construction on ice and permafrost: foundations, roads and airfields*; - Hydraulics in cold regions*; *) may not be covered depending on external circumstances	
Study Goals	On completion of this module, the student will be able to: LO1. Classify cold region structures and developments. [ILO 5, 10] LO2. Predict expected ice features and ice regimes in different cold regions. [ILO 2, 5] LO3. Account for material behaviour in cold environments. [ILO 1, 4, 5] LO4. Analyse the physics, mechanics and thermodynamics of ice. [ILO 4, 10] LO5. Assess static ice-induced loads on port, coastal, marine and offshore structures. [ILO 2, 3, 4] LO6. Analyse and select optimal structure types and structural designs for performance in cold regions depending on local conditions. [ILO 1, 2, 3, 4, 7 8] LO7. Characterize and model dynamic ice-structure interaction. [ILO 2, 4]	
Education Method	Lectures, discussion of examples during practicals and exercises, as well as formative assignments.	
Assessment	Several formative group assignments on: 1. Ice material: Growth & Thermodynamics 2. Ice-structure interaction 3. Ice dynamics These assignments do not count towards the final grade, but serve as a means to test your knowledge and receive feedback on your understanding. Note furthermore that the size of the groups depends on the number of students in the course and submission deadlines for the group assignments are communicated through Brightspace. The course is concluded with an individual written open-book exam, where the pass grade is a minimum of 5,8.	
Special Information	This course is a new and updated version of the former course OE44115 Arctic Engineering.	
Expected prior Knowledge	Structural Mechanics / Dynamics (BSc)	
Academic Skills	Critical thinking, knowledge, design of experiments, design, writing, reporting	
Literature & Study Materials	Chapters from the ASCE Handbook for Cold Regions Engineering, Lecture notes on offshore engineering in Cold Regions, lecture slides, selection of scientific publications.	
Judgement	See Assessment	
Permitted Materials during Exam	For more information on grading, see article 14 in the Rules and Guidelines (RGBE): https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations Open book exam	
Colleggerama	Yes	

MT44030	Torsion and Shear in Marine Structures	5
Responsible Instructor	L. Pahlavan	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The course is about torsion of ship and offshore structures. The loading of marine structures consist of {normal, shear} force and {bending, torsion} moment contributions. Torsion becomes particularly important in case of staggered cargo loading conditions and operation in quartering seas, introducing a shear {strain, stress}- and even normal {strain, stress} response in the beam (like) structural members. To be able to establish and calculate the marine structure torsion loading induced response, course subjects include: Free warping torsion: beam {stiffness, strength} characteristics for: - solid cross-sections (Saint-Venant theory) - hollow cross-sections: thin walled open (membrane analogy) - hollow cross-sections: thick walled closed - hollow cross-sections: thin walled closed; single-cell (Bredt approximation) - hollow cross-sections: thin walled closed; multi-cell - hollow cross-sections: thin walled mixed (hybrid). Free warping torsion: beam displacements for - solid cross-sections - hollow cross-sections. Constrained warping torsion: beam {shear, normal} stress for - hollow cross-sections: thin walled open - hollow cross-sections: thin walled closed. Generalised (beam) torsion equation: - hollow cross-sections: thin walled open - hollow cross-sections: thin walled closed.	
Study Goals	Theory (fundamentals): - to model free warping torsion of beam (like) structural members mathematically and {establish, calculate} the {stiffness, strength} characteristics as well as displacements for {solid, hollow} cross-sections - to model constrained warping torsion of beam (like) structural members mathematically and {establish, calculate} the introduced {shear, normal} stresses for hollow cross-sections - to establish the generalised (beam) torsion equation for hollow cross-sections Application (engineering): - to identify the torsion load carrying members (efficiency) in marine structures at {local, global} level - to {establish, calculate} the torsion loading induced response of marine structures for analysis and design purposes.	
Education Method	Lectures (using peer education) and assignments	
Assessment	Assignment reports (80%) and oral exam (20%)	
Department	3mE Department Maritime & Transport Technology	

MT44090	Fluid - Structure Interaction in Marine Structures	5
Responsible Instructor	Dr.ir. J.H. den Besten	
Instructor	Dr. A. Grammatikopoulos	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	In order to define the topology and scantlings of body shape and structural members a fluid and structural dynamic analysis is required. The fluid dynamics analysis results include rigid body displacements and loading; input for the structural dynamics analysis, providing the response: body deflections and {strain, stress, energy}. In case the body deflections are small in comparison to the rigid body displacements, a one-way coupling (a hydro-structural analysis) is sufficient. However, if the body deflections become larger, the fluid dynamics; the loading, becomes non-negligibly affected in turn meaning a two-way coupling (a hydro-elastic analysis) is mandatory. Although the rigid body displacements hardly change, the response consequences can be significant as well as the limit state (i.e. buckling & ultimate strength, fatigue & fracture) effects. To be able to establish the hydro-structural and hydro-elastic response and limit state characteristics of marine (i.e. fixed and floating offshore) structures, course content includes: Fluid structure interaction phenomena, i.e. hydro-structural and hydro-elastic response aspects; one-way/two-way coupling formulations, for: 1. continuous wave loading: a. local, regional and global (whipping) structural response considerations b. frequency domain solutions c. testing (measurement techniques) 2. transient wave loading: impact (external: slamming, internal: sloshing) a. Von Karman and Wagner slamming model, including inertia and free vibration stage characteristics in case of wedge and cylinder water entry as well as wave impact b. local, regional and global (springing) structural response considerations c. time domain solutions: implicit/explicit numerical integration using Newmark and 4th order Runge-Kutta formulations d. testing (measurement techniques) Fluid dynamics: define and select body shape topology and scantlings using rigid body displacements and loading based on: 1. hydrodynamic boundary value problem formulations 2. local pressure- and global force/moment distributions Structural dynamics: define and select topology and scantlings of structural members using body deflections and stress/strain/energy response based on: 1. Euler/Timoshenko beam element formulations 2. modal decomposition (frequency; natural (vibration) and wave (flexible) mode) 3. dry/wet characteristics 4. PS-SB symmetry (MB,V) and anti-symmetry (MB,H, MT) components Hydro-structural and hydro- elastic design criteria: loading and structure natural period aspects at local, regional and global level Limit state consequences: buckling & ultimate strength and fatigue & fracture Scaling (for global hydro-structural and hydro-elastic response testing purposes): 1. mass and stiffness properties and distributions 2. dry/wet natural frequency and natural (vibration) and wave (flexible) mode shape characteristics 3. backbone cross-section definition (hull girder similarity)	
Study Goals	Theory (fundamentals): to identify and describe the fluid structure interaction physics and phenomena for continuous/transient wave loading and vortex induced vibrations at local/regional/global marine structural response level to establish hydrodynamic boundary value problem formulations as well as local pressure- and global force/moment distributions in order to define and select body shape topology and scantlings to establish Euler/Timoshenko beam formulations, modal decomposition, dry/wet characteristics and symmetry/anti-symmetry components in order to define and select topology and scantlings of structural members to model hydro-structural and hydro-elastic phenomena for continuous/transient wave loading and vortex induced vibrations mathematically based on different formulations developed over time Validation (experiments): to evaluate dedicated fluid structure interaction testing techniques to apply scaling laws for global hydro-structural and hydro-elastic response testing purposes w.r.t. mass and stiffness properties and distributions as well as dry/wet characteristics in order to establish backbone hull girder cross-section similarity to analyse and process measurement data and establish hydro-structural and hydro-elastic design criteria Application (engineering): to identify and classify if a hydro-structural or hydro-elastic structural response calculation is sufficient	

Education Method	to calculate the hydro-structural and hydro-elastic response for analysis and design purposes in order to establish and improve the marine structure integrity & longevity; limit state consequences
Assessment	Lectures (blended)
Remarks	For qualification purposes assignments will be provided. Analysis, design and optimisation tasks to be performed require use of MATLAB, MAPLE and ANSYS. The assignment reports will be assessed and graded. For evaluation purposes, an appointment will be scheduled in order to discuss the results and to verify up to what extent the course objectives are achieved.
Department	Course is programmed as a 2nd year MSc course.
	3mE Department Maritime & Transport Technology

OE44145	Research Exercise	6
Responsible Instructor	Dr.ir. P. van der Male	
Instructor	Prof.dr. A. Metrikine	
Contact Hours / Week x/x/x/x	x/0/0/0 (exercise is part of the 2nd MSc year)	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Assessment	Appointment	
Remarks	Old course code: OE4611-13	
Department	3mE Department Maritime & Transport Technology	

RO47001	Robot Dynamics & Control	5
Responsible Instructor	L. Peternel	
Instructor	J.M. Prendergast	
Instructor	B. Shyrokau	
Contact Hours / Week x/x/x/x	4.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides the basics for deriving kinematic/dynamical models of a robot (such as a factory robotic arm, mobile robot, etc.) and for designing corresponding robot controllers. Students will learn how to derive the equations of motion and basic control equations themselves, and they will also learn how to use existing frameworks to do the same. THEORY SUBJECTS Introduction & Fundamentals -Course layout, instructors, grade scheme, etc. -Importance of the course and the big picture -Connection to real-world -Basic concepts/terms in robotics -Methods to describe robot pose -Numerical integration Recap of Control Theory -State-space model -Root locus -Bode plot -PID control -Concept of LQR and MPC control -Reference to online materials Endpoint Control for Robotic Manipulators -Big picture + motivation -Difference between position and torque-controlled robots -Endpoint position control (with a PID controller) -Endpoint impedance control (for position tracking tasks) -Endpoint force control (for force tracking tasks) Robot Arm Kinematics -Deriving forward kinematics and Jacobian (for a 2-DoF planar arm) -Transformations between endpoint and joint space using Jacobian -Kinematic singularities and their solution -Kinematic redundancy and task priority control (for a 4-DoF planar arm) -Kinematics of parallel robots and differences compared to serial robots Robot Arm Dynamics -Rigid-body dynamics model -Deriving dynamics model with Lagrange method for 2-DoF planar arm -Using the dynamics model in the control -Dynamic task priority control Kinematics & Dynamics of Mobile Robot / Automated Vehicle -Forward kinematic models -Wheel kinematic constraints -Robot kinematic constraints -Non-holonomic system -Derivation of linear bicycle model -Steady-state analysis Dynamics & Control of Mobile Robot / Automated Vehicle -Transient analysis -Frequency response -Path-following control 3D Robot Arm -Deriving 3D kinematics using the Denavit-Hartenberg convention -3D trajectory generation Planning & Motion Control -Orientation control -Stability analysis of the controller PRACTICAL ASSIGNMENTS (individual) Assignment 1: The goal of this assignment is to learn how to control a 2-DoF robot arm in Python to follow a prescribed trajectory using kinematic and dynamic equations. Assignment 2: The goal of this assignment is to learn how to implement a linear bicycle model, and calculate and discuss primary handling characteristics, poles, stability analysis, and feedback control (PID). Assignment 3: The goal of this assignment is to learn how to control a multi-DoF robotic manipulator (3D kinematics, 3D trajectory, orientation control).	
Study Goals	After completing this course, the students are able to:	

Education Method	-Derive the equations of motion for different types of robots -Create joint-level and endpoint-level controllers (including trajectory following) -Implement a simulation model in Python or Matlab -Analyse the designed control systems and reflect on the design choices -Lectures & interactive sessions (4 hours per week) -Practical assignments (homework)
Literature and Study Materials	Note: While attending lectures & interactive sessions is not mandatory, it is highly recommended. Assignments are not strictly mandatory, however, if you do not submit them you will lose the corresponding percentage of the final grade. That might mean that even if you pass the exam, you might not have a high enough final grade to pass the course. Study materials -Lecture slides -Reader (available in PDF format and becomes available during the course) Robotics literature (either one) -Lynch, K. M., & Park, F. C. (2017). Modern robotics. Cambridge University Press. -Craig, J. J. (2017). Introduction to robotics: mechanics and control. Pearson Education. -Siciliano, B., Sciavicco, L., Villani, L., & Oriolo, G. (2010). Robotics: modelling, planning and control. Springer Science & Business Media. Control theory literature (either one) -Ogata, K. (2010). Modern control engineering. Prentice hall. -Dorf, R. C., & Bishop, R. H. (2001). Modern Control Systems: Solutions Manual (Vol. 12). Prentice Hall. Useful toolboxes -Corke, Peter. Robotics Toolbox for Python. -Foote, Tully. "tf: The transform library." 2013 IEEE Conference on Technologies for Practical Robot Applications (TePRA). IEEE, 2013.
Prerequisites	-Linear algebra (including matrix and vector operations) -Calculus -Linear system theory at the BSc level (e.g., simple differential equations, Laplace transforms) -Basic programming experience at the BSc level (Python/Matlab)
Assessment	Number of assessments: -Four (3 assignments, 1 exam), all based on individual work -The exam will be a computer/digital exam in ANS on campus. In the case of COVID-19 restrictions, the exam will be held online (still in ANS). Grade calculation: -If and only if the exam grade is 5.5 or higher, the grade will consist of 30% of the average score of the assignments and 70% of the exam grade. The final grade must be 6.0 or higher to pass the course. Assignment submission lateness criteria: <1 hour: -1 point <1 day: -2 points <2 days: -4 points <3 days: -6 points >3 days: no points given
Department Contact	Note: Assignments and exam grades are only valid for the academic year they were obtained in. 3mE Department Cognitive Robotics For general course inquiries: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 1 and related content: Luka Peternel (L.Peternel@tudelft.nl) For Assignment 2 and related content: Barys Shyrokau (B.Shyrokau@tudelft.nl) For Assignment 3 and related content: Micah Prendergast (J.M.Prendergast@tudelft.nl)

RO47002	Machine Learning for Robotics	5
Responsible Instructor	Dr. J. Kober	
Instructor	Dr. H. Caesar	
Contact Hours / Week x/x/x/x	6.0.0.0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course provides a broad overview of machine learning techniques and their practical application in robotics. Intro to Machine Learning and python - Machine Learning in Robotics - Understanding the goal of machine learning, fundamental problems, high-level overview. Visualizing your data - Supervised vs Unsupervised vs Reinforcement learning - Model-driven vs Data-driven: White box, Gray box, Black box - Prior knowledge vs Unstructured. Feature extraction, linear regression + Deep Learning. Interpretability Hands-on Machine Learning - Example machine learning project in Robotics - Binary classification, decision boundaries, using logistic regression/SVM (one parameter) - How to perform a classification experiment. Dataset splitting, learning curves, metrics, comparing results Regression & Data collection - Regression and Data Collection in Robotics - Regression methods, least squares fitting - Overfitting, cross validation, regularization - Collecting (noisy) data, labelling data, outliers - High-dimensional data, data augmentation - Hyper-parameter optimization (grid search vs random search) Classification - Classification in Robotics - Parametric vs Non-parametric classifiers - Logistic regression - Decision tree, forest - Bayesian classification: Bayes' rule, naive Bayes, Gaussian Mixture Model - k-nearest neighbour - SVM, kernel-SVM, dual problem - Multi-class classification, metrics (confusion matrix), class imbalance Unsupervised Learning - Unsupervised Learning in Robotics - Clustering: K-means, Gaussian Mixture Model, DBSCAN - Dimensionality reduction: PCA, Local Linear Embedding (LLE) Neural Networks - Neural Networks in Robotics - Multi-Layer Perceptron, gradient descent - Neural Networks, and deep learning Advanced machine learning - Outlook: Vanishing gradient problem, Dropout, Optimizers, Data augmentation, AutoEncoder - Outlook: Time Series (RNN, LSTM) - Outlook: Reinforcement learning	
Study Goals	Knowledge, insight, judgement and skills: After completing this course, students will be able to - explain the role of ML in robotics, and describe possible use cases, and when not to use ML - explain the main concepts in ML (overfitting, training and validation, classification vs regression, cross validation, regularization, parametrics vs. non-parametrics, supervised/unsupervised/reinforcement learning) - explain the principles of well-established ML methods (e.g. Logistic regression, SVM, RFs, Neural Networks) - collect and preprocess data (normalization, feature selection, augmentation, splitting) for ML - analyze a ML problem, consider available data and computational resources, and select the appropriate ML methods to apply - perform experiments, evaluate the results, and draw sound conclusions - present the findings of a ML experiment clearly in a report using graphs and scientific English - write python code to visualize data and implement ML methods using popular python frameworks, such as pandas, numpy, matplotlib, jupyter notebook, sklearn, keras Transferable and interpersonal skills: After completing this course, students will be able to - collaborate and coordinate with a lab partner on coding and reporting tasks	
Education Method	Contact hours - Flipped classroom: pre-recorded lectures (2 hours per week) + classroom discussion (2 hours per week) - Q&A on practica (2 hours per week) Formative feedback - Quizzes -- Individual -- On material of previous lecture(s) -- Via BrightSpace at set time slots - Practica -- Pairwise collaboration -- Submission via BrightSpace -- Counts for knock-out criteria -- No late submission -- No resit Deadlines - For each practicum, the deadline is approx. one week after the practicum is published.	
Literature and Study Materials	Study materials - Slides (released shortly before lectures) - Book: "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems" 2nd edition, Aurélien Géron	

Prerequisites	Suggested literature - "Pattern Recognition and Machine Learning", C. Bishop - "The Elements of Statistical Learning: Data Mining, Inference, and Prediction", Hastie, Tibshirani, Friedman TU Delft BSc Mechanical Engineering curriculum, or equivalent. Especially familiarity with - Programming in python - Linear Algebra - Probability theory
Assessment	Final assignment (30%) - Pairwise collaboration - Late submission: -1 grade point per day - No resit - Deadline: Will be released after the 6th practicum, giving the students approx. two weeks to read the assignment, collect the necessary data, perform their experiments, and write their report Exam (70%) - Individual - Written closed-book exam - Exam in Q1, resit in Q2 Knock-out criteria - At least 4 practicum grades need to be a pass (of current or previous academic year) - Minimum grade for final assignment is a 5 (of current or previous academic year) - Minimum grade for the exam is a 5 - Final grade minimum is a 6
Department	3mE Department Cognitive Robotics

SC42001	Control System Design	5
Responsible Instructor	Dr.ir. A.J.J. van den Boom	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	State-space description of single-input, single-output linear dynamic systems, interconnections, block diagrams. Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation. Dynamic response, relation to modes, the matrix exponential. Realization of transfer function models by state space descriptions, coordinate changes, canonical forms. Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback. Application of LQ regulator. Observability, detectability, unobservable modes, state-estimation observer design. Output feedback synthesis and separation principle. Reference signal modeling, integral action for zero steady-state error; Analysis in robust stability and robust performance. Basics of model predictive control (MPC). Different model-structures. Prediction models in state-space setting. Standard predictive control scheme. Relation standard form with GPC, LQPC and other predictive control schemes. Solution of the standard predictive control problem. Stability of MPC.	
Study Goals	By taking this course, the student - will be able to master the introduced theoretical concepts in systems theory and feedback control design - will be able to practically apply these concepts to design projects and tasks - and will be able to relate the learned concepts and techniques to other more specialized ones, to potentially integrate them by taking adjacent courses. - will be able to translate a predictive control problem into a standards setting and solve the predictive control problem. More specifically, the student will be able to: - Translate differential equation models into state-space and transfer function descriptions - Rationalize differences between state-space and transfer function approaches - Linearize a system, determine its equilibrium points, analyze directly its local stability, leverage Lyapunov theory to study general stability properties - Describe the effect of eigenvalue/pole locations to the dynamic system response in time/frequency domain. Contrast step and impulse responses. Analyze transients and steady-state - Investigate model controllability. Formulate and apply the procedure of pole-placement by state-feedback, as well as LQ optimal state-feedback control - Derive observability properties. Formulate and apply the procedure of state estimation and build converging observers - Formulate the separation principle and employ it for the design of output feedback - Build reference models and achieve zero steady-state error using integral control. - set up a predictive control problem. - Solve the standard predictive control problem; - Master the main analytical details in stability proofs of predictive control schemes;	
Education Method	Lectures 4/0/0/0	
Literature and Study Materials	Textbook (its use is strongly recommended): K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers; second edition; Princeton University Press, Princeton and Oxford, 2009 Available online for download: http://www.cds.caltech.edu/~murray/books/AM08/pdf/fbs-public_24Jul2020.pdf The link for downloading the lecture notes for model predictive control will be available in the first quarter.	
Assessment	Written examination. Due to COVID-19 the exam may be an online exam. Disclaimer: information may change depending on the developments around the coronavirus.	
Department	3mE Department Delft Center for Systems and Control	

SC42015	Control Theory	6
Responsible Instructor	T. Keviczky	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Students following this course are expected to have a strong background in the following topics: - Linear algebra (basic concepts up to generalized eigenvectors and Jordan canonical form) - Calculus (smoothness, Jacobian, gradient) - Differential equations (existence, uniqueness, Laplace transform) - Classical control (transfer functions, block diagram algebra, Bode/Nyquist, etc.)	
Course Contents	- State-space description of multivariable linear dynamic systems, interconnections, block diagrams - Linearization, equilibria, stability, Lyapunov functions and the Lyapunov equation - Dynamic response, relation to modes, the matrix exponential and the variation-of-constants formula - Realization of transfer matrix models by state space descriptions, coordinate changes, normal forms - Controllability, stabilizability, uncontrollable modes and pole-placement by state-feedback - LQ regulator, robustness properties, algebraic Riccati equations - Observability, detectability, unobservable modes, state-estimation observer design - Output feedback synthesis (one- and two-degrees of freedom) and separation principle - Disturbance and reference signal modeling, the internal model principle	
Study Goals	The student is able to apply the developed tools both to theoretical questions and to simulation-based controller design projects. More specifically, the student must be able to: - Translate differential equation models into state-space and transfer matrix descriptions - Linearize a system, determine equilibrium points and analyze local stability - Describe the effect of pole locations to the dynamic system response in time- and frequency-domain - Verify controllability, stabilizability, observability, detectability, minimality of realizations - Sketch the relevance of normal forms and their role for controller design and model reduction - Describe the procedure and purpose of pole-placement by state-feedback and apply it - Apply LQ optimal state-feedback control and analyze the controlled system - Reproduce how to solve Riccati equations and describe the solution properties - Explain the relevance of state estimation and build converging observers - Apply the separation principle for systematic 1dof and 2dof output-feedback controller design - Build disturbance and reference models and apply the internal model principle	
Education Method	Lectures and Exercise Sessions	
Computer Use	The exercises will be partially based on Matlab in order to train the use of modern computational tools for model-based control system design.	
Course Relations	This course is intended primarily for MSc students that are enrolled in the Systems & Control MSc program! For students who are interested in Systems & Control within their own MSc program, the SC42001 Control System Design course is recommended instead. There is a large overlap between the two courses in terms of topics, but SC42001 emphasizes broader concepts of control, it is more application-oriented, and does not discuss the technical proofs and mathematical background so extensively.	
Literature and Study Materials	B. Friedland, Control System Design: An Introduction to State-space Methods. Dover Publications, 2005 K.J. Astrom, R.M. Murray, Feedback Systems: An Introduction for Scientists and Engineers, Princeton University Press, Princeton and Oxford, 2009 http://www.cds.caltech.edu/~murray/amwiki/index.php?title=Main_Page	
Assessment	Written mid-term examination (15% grade) and written final examination (85% grade) will determine the total final grade. The mid-term result is only valid for the academic year when it was obtained. For the resit examination there will be a written examination (100% grade) for which the mid-term result will not count. If on-campus exams will not be possible, then all exams will be organized online (via Ans).	
Department	3mE Department Delft Center for Systems and Control	

SC42056	Optimization for Systems and Control	3
Responsible Instructor	Prof.dr.ir. B.H.K. De Schutter	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	basic experience with Matlab	
Course Contents	Essentially, almost all engineering problems are optimization problems. If a civil engineer designs a bridge, then one of the main objectives is to obtain the cheapest design or the design that can be implemented most rapidly, where of course several specifications and constraints such as size, strength, safety, etc. have to be taken into account. When developing a new type of engine, we look for the most economical design, the cheapest design, or the design with the highest performance. A process engineer wants a production unit to deliver a final product of maximal quality, with minimal expenditure of energy or with maximal output flow. When composing a portfolio, a financial engineer tries to maximize the expected profits, subject to the given risk constraints. So we encounter optimization problems in almost every engineering field. How can we solve such an optimization problem? That is the topic that will be addressed in this course. We will in particular discuss several classes of optimization problems and next focus on how to select most efficient numerical algorithms to solve a given optimization problem. The examples and case studies of this course are primarily oriented towards systems and control. More specifically, the following topics will be treated: * Linear Programming * Quadratic Programming * Nonlinear Optimization * Constraints in Nonlinear Optimization * Convex Optimization * Global Optimization * Matlab Optimization Toolbox * Multi-Objective Optimization * Integer Optimization	
Study Goals	After this course the students should be able to select the most efficient and best suited optimization algorithm for a given optimization problem. They should have insight into the basic operation of some of the most important and relevant optimization algorithms. They should be able to reduce the complexity of the problem using simplifications and/or reformulations so as to augment the efficiency of the solution approach. Finally, they should be able to solve the resulting optimization problem in practice.	
Education Method	lectures + 2 assignments Depending on the corona measures, the lectures will be taught on-campus or online.	
Assessment	Attending the lectures is not mandatory, provided the recorded lectures and/or lecture materials are the processed via self-study. exam (individual, counts for 75% of the final marks) + 2 assignments (group assignments, assessed through written reports, count for 25% of the final marks) Important: Partial marks for the exam or the assignments do not carry over from one academic year to the next. Depending on the of measures resulting from COVID-19, the prescribed exam will either be an open-book online exam, or if the opportunity arises for examinations on campus, a closed-book written exam (fully closed-book, no calculators nor formulas sheets allowed).	
Permitted Materials during Tests	none	
Department	3mE Department Delft Center for Systems and Control	
Contact	questions are preferably asked via the Discussion forum on Brightspace or (for questions on the material of the lectures) during the online or on-campus lectures	

SET3055	Economics and Regulations of Sustainable Energy Systems	4
Responsible Instructor	Dr. S.T.H. Storm	
Instructor	Prof.dr.ir. L.J. de Vries	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	None	
Course Contents	This course is a general introduction to energy economics. In this course, you learn the essentials of how actors in the energy system make economic decisions, when markets work well and when they may fail, and what options the government has to remedy market failures. Specifically, the following topics are addressed: Introduction to energy (electricity) markets: perfect competition; oligopoly Rational (and irrational) market behaviour; ABM ? Investment decision making; LCOE Opportunity costing / discounting; NPV; IRR; discount rate Markets, (perfect) competition and market failure (esp. externalities & lock-in) Private costs, external cost and social cost; empirical evidence (related to energy) LCOE; discounting; private discount rate; IRR and NPV Social cost-benefit analysis; social discount rate	
Study Goals	After this course students are able to: evaluate and analyse the performance of energy (electricity) markets; identify cases of market failures and design remedial action to rectify these market failures; estimate the LCOE and calculate the NPV and IRR of an electricity investment project; contrast private and social costs of (renewable) electricity generation; assess technological lock-in in energy generation; and appraise the role of public policy in energy markets and in energy technology innovation.	
Education Method	This course consists of lectures. Parallel to the course a simulation game will be played by the students.	
Literature and Study Materials	To be announced.	
Assessment	The final grade of the course consists of the following components: - Individual computer exam (80%) - Report on the simulation game (20%) Students should score at least 6 for both parts. All marks will be expressed on a scale of 1 to 10. Final grade calculation: $(0.8 * \text{computer exam}) + (0.2 * \text{report})$ In case of an insufficient result, repair opportunities may be offered in accordance with TER Implementation Regulations Art 5, sub 5., for: - Computer exam: resit computer exam - Report on the simulation: resubmission of the report Disclaimer: information may change depending on unforeseen circumstances or measures (see: TER Art 29, sub 4)	
Co-Instructor	G.O. Ndubuisi	

TPM404B	Technology Entrepreneurship and Internationalization	5
Module Manager	H. Khodaei	
Instructor	F. Delgado Medina	
Instructor	H. Khodaei	
Co-responsible for assignments	Dr.ing. V.E. Scholten	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course offers theoretical fundamentals of international entrepreneurship. You create a case study of an existing start-up (born global) analysing its entrepreneurial journey and how it fits the broad technology development process. This course helps you prepare for a wide array of career paths, from starting or joining an international start-up to academic research on the topics of International entrepreneurship. Throughout the course you will develop the skills to analyse the emergence of innovative business opportunities and how these are exploited. Using proven content, methods, and models for new venture opportunity assessment and analysis, you will learn how to analyze the commercial potential of implementing new technologies either by analyzing social challenges or analyzing the value of technology for various user applications. Innovation, opportunity recognition, strategic decision making, business modelling, internationalization marketing and growth stages of high-tech start-ups are core elements of the course.	
Study Goals	Upon finishing the course you will be able to: LO 1 Analyse and apply the innovation process within startup firms and compare it to established firms. LO 2 Describe and apply opportunity identification process based on opportunity analysis canvas. LO 3 Identify and analysis international opportunities across different markets. LO 4 Describe and apply the born-global start-up growth process. LO 5 Demonstrate understanding of the entrepreneurial ecosystem, the main stakeholders and their role in the start-up process. LO 6 Explain and apply methods for developing business models for high-tech start-up and analyse how they can change over time. LO 7 Describe and analysis the barriers to internationalization and evaluate the niche strategies. LO 8 Analyse an existing born-global start-up and present your findings about the dynamics of its growth.	
Education Method	The course is organized into 8 sessions (8 weeks) into 3 learning blocks with an assessment at the end of each. Each session consists of a mixture of lecturing, and discussions of cases, literature and assignments, and presentations. Assignments provide you with opportunities to integrate and apply learning from each learning block. A final presentation and final report based on a startup-case analyses provided by a team of students will close the course. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations will be facilitated in an online mode (for example through zoom or Skype). Book: Trott, P., Hartmann, D., Scholten, V., van der Duin, P., & Ortt, J. R. (2015). Managing technology entrepreneurship and innovation. New York, USA Journal articles: Will be posted in Brightspace Slides presented in class: Will be posted in Brightspace	
Assessment	The assessment is as follows: - Team: Case analyses (30%) and a final report and presentation (40%) - Individual: theory quizzes (30%) All marks will be expressed on a scale of 1-10. In case of unforeseen circumstances or measures resulting from COVID-19, the presentations and individual assessment will be facilitated in an online mode (for example through zoom or Skype)	
Elective	Yes	

TUD4040	Joint Interdisciplinary Project	15
Responsible Instructor	Prof.dr.ir. J. Hellendoorn	
Project Coordinator	Ir. B.J.E. de Bruin	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	Different, to be announced	
Course Language	English	
Summary	JIP consists of three sets of activities: 1. The project which takes place at a company. Students are responsible for the project work (collaboration, integration of different perspectives, the content quality of their work). They plan their activities via a scrum method, are challenged to realise field trips, consult with experts and realise the necessary research work. The team keeps a Scrum log, run a blog for interaction with the outside world and every team member keeps a personal development log. 2. Meeting Professionals lectures and workshops about specialised topics, by academic staff or senior professionals from the companies involved. 3. Plenary meetings and design reviews for all the teams, company coaches and academic staff. At the meetings the students present their intermediate resp. final outcomes and plans, and are provided with feedback.	
Course Contents	The aim of the Joint Interdisciplinary Project is to prepare students to contribute to solving impactful technological challenges. The projects not only demand good engineering working knowledge but also experience with interdisciplinary and systems theory, and both knowledge and mindsets of innovation and entrepreneurial behavior. The project brief is provided by renowned companies like Airbus, Arcadis, etc. Teams of interdisciplinary student teams guided by a company coach and offered academic and industry expertise, are invited to realise an innovative problem solution to a complex problem and contributing to the sustainable development goals.	
Study Goals	1. Cognitive abilities attributable to interdisciplinary learning Demonstrate the ability to engage in perspective-taking; Develop structural knowledge pertaining to the problem; Integrate knowledge and modes of thinking drawn from two or more disciplines; Produce an interdisciplinary understanding of complex problem or intellectual question. 2. Scientific and intellectual development Capable to analyze scientific and societal consequences (economic, social, cultural, environmental) of the innovation; 3. Research and design capabilities Demonstrate engineering skills: technical skills, interpreting results, creativity, usability for company/institute; Demonstrate that they are capable to independently apply relevant theory and/or knowledge to research and/or design; 4. Collaboration and communication in an interdisciplinary team Demonstrate behavioral competences and skills: taking initiative, responsibility, showing communication skills, independency, collaboration and the ability to respect different disciplines and adapt to different cultures); Show ability to write a technical report: structured/consistent, language proficient, with correct use of literature/references, use of figures/tables/equations, and has a concise format (30 pages); Present work performed in a structured way through an oral presentation to their peers and customer. 5. Self-adjustment and reflection capabilities Plan and control the project efficiently considering resources and methodology; Being able to reflect on personal functioning in an evaluation report: reflect on personal objectives, indicate personal strengths/weaknesses. Indicate future personal improvement, drawing conclusions for future career. Cognitive abilities attributable to interdisciplinary learning: The ability to integrate (scientific and practical technological) knowledge from different disciplines to solve complex problems Scientific and intellectual development The capacity to evaluate the ethical, scientific and societal consequences of the proposed innovation Research and design capabilities The ability to create reasonable and relevant research or design , according to the academic standards of the involved disciplines Collaboration and communication in an interdisciplinary team Demonstrate behavioural competences and skills relevant for teamwork and effective communication with different stakeholders Self-adjustment and reflection capabilities To carry out regular reflections on professional and personal development and being able to improve upon those reflections Understand contemporary and societal issues in their work.	
Education Method	Full-time project work in an interdisciplinary team of about five students. The project work is interspersed with some just in time seminars and workshops about specialized subjects, methods or practical situations that play an essential role in interdisciplinary project work.	
Assessment	The assessment criteria pertain to the process, product and presentation and are assessed at an individual and team level, during 3 review sessions at three levels of accomplishment (each review assesses at a different level): Interdisciplinary work Scientific reasoning and ethical mindset Innovation process Presentation and communication Academic staff, supported by the input from company coaches, grade the students during the Problem Statement, the Midterm and the Final Review. The final group mark for JIP is differentiated per person and is based on: The team presentations - 30% The Problem definition, progress report and final report 30% The individual contribution to the team 20% The final individual reflection report 5% The Blog 5%	
Remarks	Project Outcomes (product, model, system) are presented at: 1. Problem Statement Review 2. Midterm meeting 3. Final Review 4. Symposium presentation	

S.M.S. Alhaddad

Department	Offshore and Dredging Engineering
Telephone	+31 15 27 83675
Room	34.B-2-230

Dr. K. Anupam

Unit	Civiele Techniek & Geowetensch
Department	Pavement Engineering
Telephone	+31 15 27 82394
Room	23.HG 6.60

Dr.ir. J.H. den Besten

Unit	Mech, Maritime & Materials Eng
Department	Ship and Offshore Structures
Telephone	+31 15 27 86507
Room	34.B-1-290

Dr.ir. A.J.J. van den Boom

Unit	Mech, Maritime & Materials Eng
Department	Team Ton van den Boom
Telephone	+31 15 27 84052
Room	34.C-3-180

Ir. B.J.E. de Bruin

Department	Cognitive Robotics
Department	Cognitive Robotics

Dr. H. Caesar

Department	Intelligent Vehicles
Telephone	+31 15 27 82419

J.O. Colomes Gene

Department	Offshore Engineering
Telephone	+31 15 27 83461
Room	23.HG 3.65

A. Coraddu

Department	Ship Design, Production and Operations
Telephone	+31 15 27 87809
Room	34.B-4-180

Prof.dr.ir. B.H.K. De Schutter

Unit	Mech, Maritime & Materials Eng
Department	Delft Center for Systems and Control
Telephone	+31 15 27 85113

Dr.ir. D.A.M. De Tavernier

Department	Arts & Crafts
Telephone	+31 15 27 82606
Unit	Luchtvaart- & Ruimtevaarttechn
Department	Wind Energy
Telephone	+31 15 27 82606

F. Delgado Medina

Department	Delft Centre for Entrepreneurship
-------------------	-----------------------------------

Dr. J.M. Duran

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek

Telephone	+31 15 27 87879
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Dr. A. Grammatikopoulos

Department	Ship and Offshore Structures
-------------------	------------------------------

Prof.dr.ir. J. Hellendoorn

Department	College van Bestuur
-------------------	---------------------

Telephone Room	+31 15 27 89007 34.F-2-440
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Unit Department	Mech, Maritime & Materials Eng Cognitive Robotics
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Telephone Room	+31 15 27 89007 34.F-2-440
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Dr.ir. R.L.J. Helmons

Unit Department	Mech, Maritime & Materials Eng Offshore and Dredging Engineering
------------------------	---

Telephone	+31 15 27 84746
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Dr.ir. H. Hendrikse

Unit Department	Civiele Techniek & Geowetensch Offshore Engineering
------------------------	--

Telephone Room	+31 15 27 88223 23.HG 3.71
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Ir. J.S. Hoving

Unit Department	Civiele Techniek & Geowetensch Offshore Engineering
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Telephone Room	+31 15 27 85723 23.HG 3.61.1
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Dr.ir. A. Jarquin Laguna

Unit Department	Civiele Techniek & Geowetensch Offshore and Dredging Engineering
------------------------	---

Telephone	+31 15 27 83297
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Dr.ir. G.H. Keetels

Unit Department	Mech, Maritime & Materials Eng Offshore and Dredging Engineering
------------------------	---

Telephone	+31 15 27 84057
------------------	-----------------

Dr. E. Kementzetzidis

Unit Department	Civiele Techniek & Geowetensch Offshore Engineering
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Room	23.HG 3.45
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T. Keviczky

Unit Department	Mech, Maritime & Materials Eng Team Tamas Keviczky
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Telephone Room	+31 15 27 82928 34.C-3-310
-----------------------	-------------------------------

H. Khodaei

Unit Department	Techniek, Bestuur & Management Delft Centre for Entrepreneurship
------------------------	---

Telephone Room	+31 15 27 82759 31.C2.130
-----------------------	------------------------------

Dr. J. Kober

Unit Department	Mech, Maritime & Materials Eng Learning & Autonomous Control
------------------------	---

Telephone	+31 15 27 85150
------------------	-----------------

Room	34.F-2-380
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Dr.ir. P. van der Male

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 83473
Room	23.HG 3.71

Prof.dr. A. Metrikine

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 84749
Room	23.HG 3.59.1

Unit	Civiele Techniek & Geowetensch
Department	Engineering Structures
Telephone	+31 15 27 84749
Room	23.HG 3.59.1

Dr.ir. P. Naaijen

Unit	Mech, Maritime & Materials Eng
Department	Ship Hydromechanics and Structures
Telephone	+31 15 27 81570

G.O. Ndubuisi

Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 88895

L. Pahlavan

Unit	Mech, Maritime & Materials Eng
Department	Ship and Offshore Structures
Telephone	+31 15 27 87682

Dr. M. Pavlovic

Unit	Civiele Techniek & Geowetensch
Department	Steel & Composite Structures
Telephone	+31 15 27 83382
Room	23.S2 2.58

L. Peternel

Unit	Mech, Maritime & Materials Eng
Department	Human-Robot Interaction
Telephone	+31 15 27 85695
Room	34.F-1-420

J.M. Prendergast

Department	Human-Robot Interaction
Telephone	+31 15 27 87944

Prof.dr.ir. C. van Rhee

Department	Rivers, Ports, Waterways and Dredging Engineering
Telephone	+31 15 27 83973
Room	34.B-2-340

Unit	Mech, Maritime & Materials Eng
Department	Offshore and Dredging Engineering
Telephone	+31 15 27 83973
Room	34.B-2-340

M. Sand

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 87885

Room	31.B4.250
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Dr. F. Santoni De Sio

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek
Telephone	+31 15 27 85141

Dr.ir. H.R. Schipper

Unit	Civiele Techniek & Geowetensch
Department	Applied Mechanics
Telephone	+31 15 27 89933

Dr.ing. V.E. Scholten

Unit	Techniek, Bestuur & Management
Department	Delft Centre for Entrepreneurship
Telephone	+31 15 27 89596
Room	31.C2.070

Dr.ing. S. Schreier

Unit	Mech, Maritime & Materials Eng
Department	Ship Hydromechanics and Structures
Telephone	+31 15 27 85562
Room	34.D-0-260

Dr. H. Seyffert

Unit	Mech, Maritime & Materials Eng
Department	Ship Hydromechanics
Telephone	+31 15 27 86366

B. Shyrokau

Unit	Mech, Maritime & Materials Eng
Department	Intelligent Vehicles
Telephone	+31 15 27 84536
Room	34.E-0-220

Ir. A.C.M. van der Stap

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Room	23.HG 3.48

Dr. S.T.H. Storm

Unit	Techniek, Bestuur & Management
Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 83548
Room	31.c.2.160

Dr.ir. A.M. Talmon

Unit	Mech, Maritime & Materials Eng
Department	Offshore and Dredging Engineering
Telephone	+31 15 27 83717
Room	34.B-2-310

A. Tsouvalas

Department	Dynamics of Structures
Telephone	+31 15 27 89225
Room	23.HG 3.61

Unit	Civiele Techniek & Geowetensch
Department	Offshore Engineering
Telephone	+31 15 27 89225
Room	23.HG 3.61

Dr.ir. A.C. Viré

Unit	Luchtvaart- & Ruimtevaarttechn
Department	Wind Energy
Telephone	+31 15 27 81385
Room	62.5.19

Prof.dr.ir. L.J. de Vries

Unit	Techniek, Bestuur & Management
Department	Energie and Industrie
Telephone	+31 15 27 81137
Room	31.B3.030

Dr. C.L. Walters

Unit	Mech, Maritime & Materials Eng
Department	Ship and Offshore Structures
Telephone	+31 15 27 83705
Room	34.B-1-310

Dr.ir. P.R. Wellens

Unit	Mech, Maritime & Materials Eng
Department	Ship Hydromechanics and Structures
Telephone	+31 15 27 88251

Ir. S.J. Zwart